

Ribbon concordances and slice obstructions: experiments and examples

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Dedicated to Cameron McA. Gordon for his impact on topology and topologists.

Abstract. There are 352.2 million prime knots in the 3-sphere with at most 19 crossings. We study which of these knots are slice, in both the smooth and topological categories. While no algorithm is known for deciding whether a given knot is slice in either setting, we are able to determine it smoothly for all but about 11,400 knots (0.003% or 1 in 30,000) and topologically for all but about 1,400 knots (0.0004% or about 1 in 250,000). In particular, we show that some 1.6 million of these knots (0.46%) are smoothly slice (in fact ribbon) and that 350.5 million are not even topologically slice (99.54%). We use a wide range of tools and techniques, and introduce several new or refined methods for probing these properties. Along the way, we produce 500,000 pairs of 0-friends, that is, pairs of distinct knots with the same 0-surgery. We discuss how our data is consistent with several important conjectures and suggests new ones, and highlight the simplest knots where sliceness remains unknown.

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1 Introduction

Recall a *knot* is the image K of a smooth embedding $S^1 \hookrightarrow S^3$. A knot K is *smoothly slice* when there is a smoothly embedded disk D^2 in the 4-ball D^4 whose boundary is K . Similarly, a knot K is *topologically slice* when it is the boundary of a locally-flat embedded disk in D^4 . Any smoothly slice knot is topologically slice, but the converse need not be the case. There are 352.2 million prime knots in the 3-sphere with at most 19 crossings [Bur]. We study which of these knots are slice, in both the smooth and topological categories. While no algorithm is known for deciding whether a given knot is slice in either setting, we are able to determine it smoothly for all but $\approx 11,400$ knots (0.003% or about 1 in 30,000) and topologically for all but ≈ 1400 knots (0.0004% or about 1 in 250,000), using a wide variety of tools and techniques. In particular, we show that some ≈ 1.6 million of these knots (0.46%) are smoothly slice and that ≈ 350.5 million are not even topologically slice (99.54%). More precisely:

1.1 Theorem. *Let $\mathcal{K}_{19}$ denote the nontrivial prime knots in the 3-sphere with at most 19 crossings, up to homeomorphism of S^3 , so that $\#\mathcal{K}_{19} = 352,152,252$. Then:*

- (1) *For $S = 1,633,786$ (0.46%), the number of smoothly slice knots in $\mathcal{K}_{19}$ is in $[S, S + 11,383]$.*
- (2) *For $T = S + 22,412 = 1,656,198$ (0.47%), the number of topologically slice knots in $\mathcal{K}_{19}$ is in $[T, T + 1429]$. There are at least 12,162 knots in $\mathcal{K}_{19}$ which are topologically slice but not smoothly slice.*

Consequently, for $R = 350,494,625$ (99.53%), the number of knots in $\mathcal{K}_{19}$ that are not topologically slice is in $[R, R + 1429]$.

1.2 Ribbon knots. A *ribbon disk* for a knot K is a smooth slice disk D with image in $D^4 \setminus \{0\} \cong S^3 \times (0, 1]$ so that the $(0, 1]$ -coordinate is a Morse function on D with no local maxes in the interior of D . A knot is *ribbon* when it has a ribbon disk. (Equivalently, a knot is ribbon when it is the boundary of a restricted kind of immersed disk $D \looparrowright S^3$ as in e.g. [Fox] or [Kaw, Definition 13.1.9].) As every ribbon knot is smoothly slice, one is led to ask:

1.3 Question [Fox]. *Are all smoothly slice knots ribbon?*

The slice-ribbon conjecture is that the answer to this question is yes. Here, all the smoothly slice knots identified in Theorem 1.1 are indeed ribbon.

1.4 Prior work. For knots with at most 12 crossings, the complete picture was finished only in 2020:

1.5 Theorem [LM, Pic]. *Of the 2977 nontrivial prime knots with at most 12 crossings, exactly 158 are smoothly slice and 159 are topologically slice. All of the smoothly slice knots are moreover ribbon.*

Several groups have worked to identify ribbon and/or slice knots with more crossings, including [Sto, Appendix A] and [GHMR]. The largest general list we are aware of is [GHMR], which identifies 1705 ribbon knots from among the 59,937 nontrivial prime knots with at most 14 crossings. We independently found the same 1705 knots as part of Theorem 1.1; see Section 2.9 for more.

For alternating knots, Owens and Swenton studied these questions at a vastly larger scale than Theorem 1.5 or [GHMR] using techniques specific to the alternating case:

1.6 Theorem [OS]. *Of the 1.2 billion nontrivial prime alternating knots with at most 21 crossings, at least $R = 664,633$ (0.05%) are ribbon and at most $R + 3276$ are smoothly slice.*

For alternating knots with at most 19 crossings, Owens and Swenton had only 86 knots whose smooth slice status was unknown. We resolved all of these in the course of proving Theorem 1.1 and moreover settled the question in the topological category:

1.7 Theorem. *Of the 51.3 million nontrivial prime alternating knots with at most 19 crossings, exactly 82,043 (0.16%) are ribbon and the rest are not topologically slice.*

1.8 Plausibly slice knots. We now outline the methods behind Theorem 1.1, starting with the basic topological obstructions from the 1960s. We call a knot *plausibly slice* when its Murasugi signature $\sigma(K) = 0$ and its Alexander polynomial $\Delta_K(t)$ satisfies the Fox-Milnor criteria, that is, there is an $f(t) \in \mathbb{Z}[t^{\pm 1}]$ with $\Delta_K(t) = f(t)f(t^{-1})$ up to a unit in $\mathbb{Z}[t^{\pm 1}]$. Any topologically slice knot is plausibly slice, see e.g. Theorems 8.18 and 8.19 of [Lic] or Corollary 12.3.13 of [Kaw]. There are 3,869,541 nontrivial plausibly slice prime knots with at most 19 crossings, which is 1.10% of the prime knots in that range; we use $\mathcal{PS}_{19}$ to denote these knots, and these are what we will study for the rest of this paper. (Algebraically slice knots are plausibly slice, and we suspect that nearly all knots in $\mathcal{PS}_{19}$ are actually algebraically slice. However, checking this is quite involved [Liv], and we are unaware of any software that handles the general case. Ad hoc methods applied to a random sample of 10,000 knots from $\mathcal{PS}_{19}$ not known to be topologically slice found that at least 99.9% are algebraically slice.)

Statistics of some basic properties of the knots in $\mathcal{PS}_{19}$ are shown in Figures 1, 2, and 3. Theorem 1.1 follows immediately from:

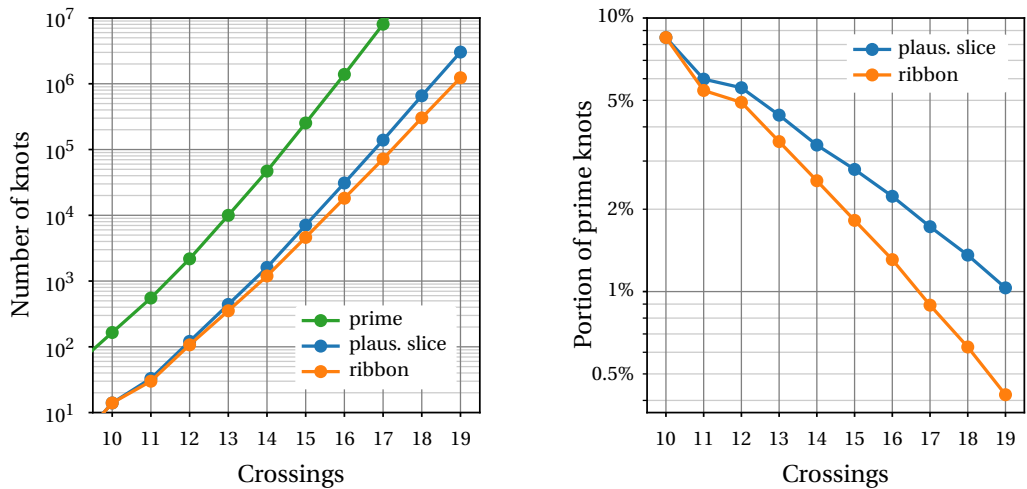


Figure 1. The number of prime plausibly slice and prime ribbon knots, as compared to all prime knots. For 20 crossings, we expect about 15 million prime plausibly slice knots and 5.5 million prime ribbon knots. (There are 1.8 billion prime knots with 20 crossings [Thi] and the plot at right suggests 0.8% are plausibly slice and 0.3% are ribbon.)

1.9 Theorem. *Of the knots in $\mathcal{PS}_{19}$, at least 1,633,786 are ribbon and at least 2,224,372 are not smoothly slice. Also, at least 1,656,198 are topologically slice and 2,211,914 are not topologically slice. This leaves 11,383 and 1429 unknowns in the smooth and topological settings respectively. There are at least 12,162 knots in $\mathcal{PS}_{19}$ that are topologically but not smoothly slice.*

Two methods handle all but 0.6% of the knots in Theorem 1.9:

- (1) All 1,633,786 ribbon knots were identified using the process detailed in Section 2. Given a knot K , we did a combinatorial search for a sequence of band moves that turns K into the unlink and so specifies a ribbon disk. As needed, we looked for such moves in many different diagrams for K ; these diagrams were generated by a more global method than just doing random Reidemeister moves. We also created a table of $\approx 12,100$ two-component ribbon links which sometimes allowed us to certify a knot is ribbon after doing a single band move.
- (2) The main slicing obstruction used was in the topological category, namely the technique of Herald, Kirk, and Livingston [HKL]. This method uses twisted Alexander polynomials to provide a computationally-efficient form of the Casson-Gordon invariants [CG2]. We introduce several refinements of this method in Sections 3.6, 3.9, 3.11, and 3.20. Collectively, such *HKL tests* obstruct 2,211,761 knots in $\mathcal{PS}_{19}$ from being topologically slice (Theorem 3.1).

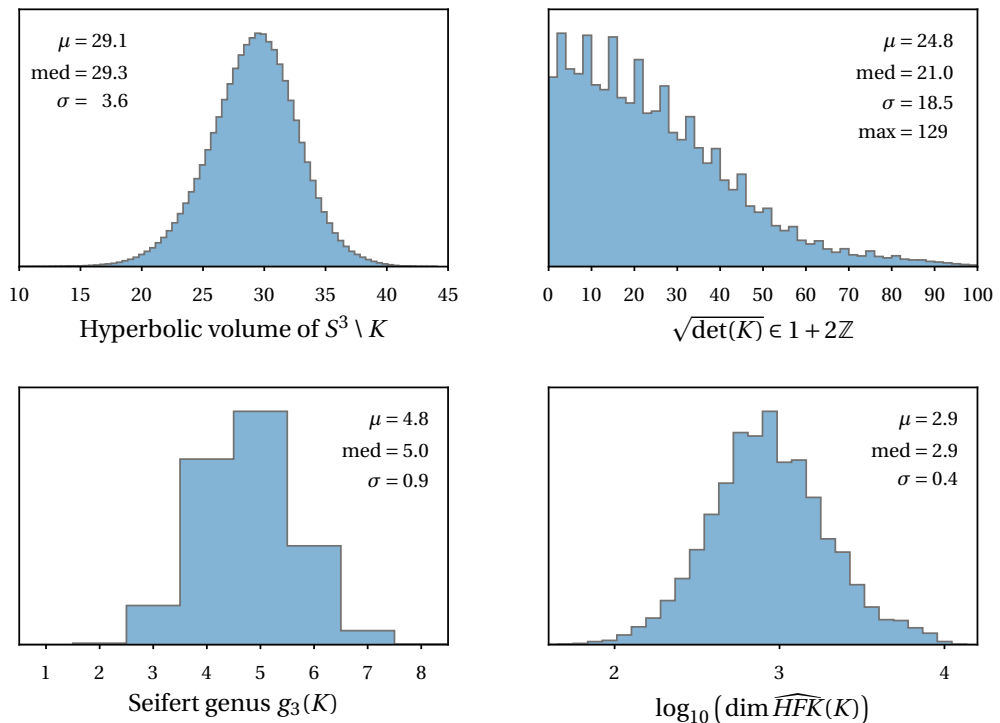


Figure 2. Some basic statistics about the 3.87 million plausibly slice knots. Here μ is the mean, med is the median, and σ is the standard deviation. The “spikes” in the plot of $\sqrt{\det(K)}$ are when that number is divisible by 3. The plot of the knot Floer homology $\widehat{HFK}$ is indistinguishable from the one of reduced Khovanov homology over $\mathbb{F}_2$ or $\mathbb{Q}$.

To handle the remaining 23,994 knots, we used the following methods:

- (3) We computed many different smooth slice obstructions from Khovanov homology, knot Floer homology, and gauge theory; see Section 4 for details, but this includes the s -invariant in Khovanov homology and several of its generalizations. Collectively, these show that 333,912 knots (8.6% of $\mathcal{PS}_{19}$) are not smoothly slice (Theorem 4.1). However, only 10,631 of these knots (0.3% of $\mathcal{PS}_{19}$) are not already covered by the HKL tests in (2).
- (4) Some 36 knots need special attention including the Conway knot and the 2-1 cable on the figure 8 knot, which are not smoothly slice by [Pic] and [DKMPS] respectively. Theorem 5.10 shows that 25 knots are not smoothly slice using the 0-friends method discussed below in Section 1.11. Theorems 6.2 and 6.3 each show a single knot is not smoothly slice. Finally, Theorem 6.6 shows nine alternating knots are not topologically slice.

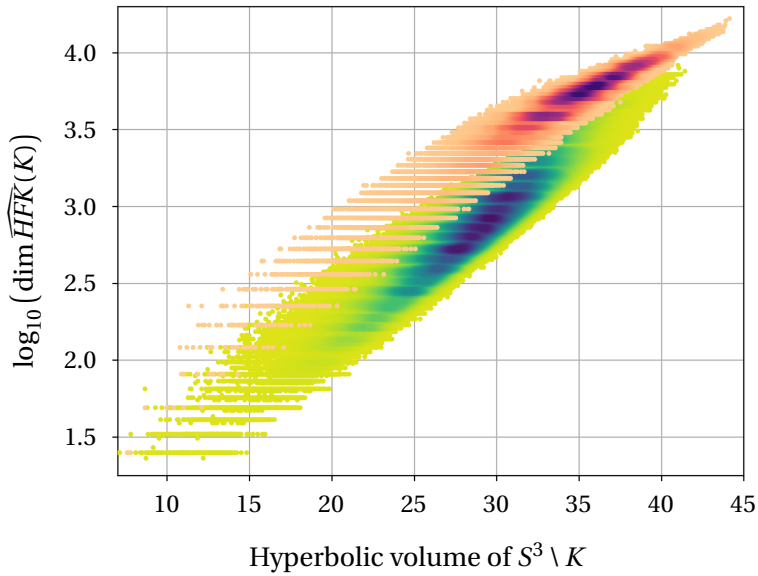
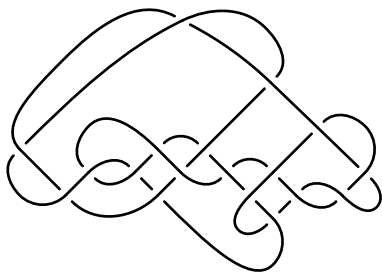


Figure 3. Comparing hyperbolic volume and the knot Floer homology $\widehat{HFK}$ for knots in $\mathcal{PS}_{19}$. Here, the alternating knots are the higher blob (peach-purple) and the non-alternating ones are the lower blob (green-blue); only 5.3% of the knots in $\mathcal{PS}_{19}$ are alternating, so this plot over-emphasizes them to show the relative behavior. This plot of $\widehat{HFK}$ is indistinguishable from the one for reduced Khovanov homology over $\mathbb{F}_2$ or $\mathbb{Q}$, so this matches the observation from [Kho, §8].

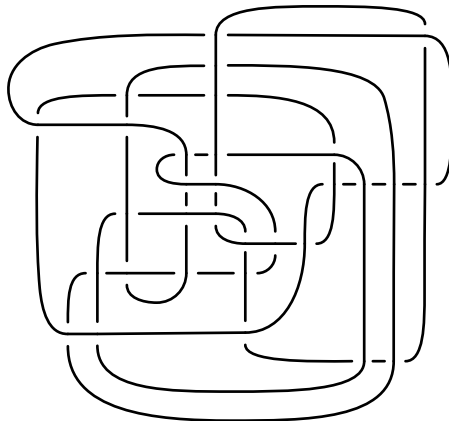
- (5) The additional 22,412 topologically slice knots were primarily identified by having $\Delta_K = 1$, see [FQ, §11.7], or by the method of Section 2 producing a ribbon concordance to a knot with $\Delta_K = 1$. One knot, $18nh_{00000601}$, was shown to be topologically slice by the 0-friend method as part of Theorem 5.14.
- (6) The final ≈ 2000 knots we dealt with in Theorem 1.9 were handled in Section 7 by using the ribbon concordances between knots in $\mathcal{PS}_{19}$ produced in Section 2. For example, there are 1672 other knots in $\mathcal{PS}_{19}$ that are ribbon concordant to the Conway knot; since the Conway knot is not smoothly slice, the same is true for those 1672 knots.

1.10 Remark. It is worth noting that the topological HKL tests of (2) were about 6.6 times as effective as all the smooth obstructions in (3). Thus, in practice it seems that one should start with HKL tests rather the smooth invariants even if one only cares about slicing in the smooth category.

Also, the 10 knots in Theorems 6.2 and 6.6 as part of (4) are the only place where we rely directly on the vast literature of slice properties of particular knots. In all other cases, our computations are at least nominally independent of any prior work.



$K = 18nh_{00000601}$ in $\mathcal{PS}_{19}$



A 0-friend K' of K

Figure 4. A pair of 0-friends. The knot K has 18 crossings and the simplest diagram for K' known has 31 crossings. This pair will appear again in Section 5.13.

1.11 0-friends. The 0-surgery on a knot K in S^3 is the unique Dehn surgery where $b_1 > 0$. A pair of knots in S^3 are *0-friends* when their 0-surgeries are homeomorphic. A secondary focus of this paper, contained in Section 5, is to study 0-friends of knots in $\mathcal{PS}_{19}$. The motivation for this is twofold. First, and most tantalisingly, if K and K' are 0-friends where K is smoothly slice and K' is not, then there is an exotic smooth structure on S^4 , disproving the smooth 4-dimensional Poincaré Conjecture; see [MP, §1]. Second, even with that conjecture unresolved, in favorable circumstances one can show that certain 0-friends must have the same smooth slice status; for example, this was used to show that the Conway knot is not smoothly slice [Pic].

There are only 101 pairs of knots in $\mathcal{PS}_{19}$ that are 0-friends (see Section 5.1), so we used the method of [DOR, §9.3] to generate more than 500,000 0-friends of the knots in $\mathcal{PS}_{19}$. Very roughly, starting with a knot K whose 0-surgery $Z_K = S^3_0(K)$ is hyperbolic, one drills out a closed geodesic in Z_K and tests whether the resulting manifold is the exterior of a knot in S^3 . When it is, one can recover a diagram for the 0-friend using [DOR]. We outline in Section 5 why hyperbolic geometry suggests this method produces most small-to-medium 0-friends of knots in $\mathcal{PS}_{19}$; a sample pair of 0-friends found is shown in Figure 4.

Using this plethora of 0-friends and the methods of [Pic] and [MP], we show 25 knots in $\mathcal{PS}_{19}$ are not smoothly slice in Theorem 5.10. Following [MP], this involves producing an RBG link (see Definition 5.7 below) for each pair of 0-friends; we introduce a new computational method to search for such links in Section 5.11, again using [DOR].

We also found four pairs where we could determine the smooth sliceness of the

0-friend but not of the original knot in $\mathcal{PS}_{19}$, leading to:

1.12 Theorem. *If $18nh_{00000601}$ is not smoothly slice, or if any of $\{16n68278, 17nh_{0010647}, 18nh_{00098198}\}$ are smoothly slice, then there is an exotic smooth 4-sphere.*

All four knots in Theorem 1.12 are topologically slice, and $18nh_{00000601}$ is even smoothly slice in a homotopy 4-ball; see Theorem 5.14.

1.13 Ribbon concordances. Suppose K_0 and K_1 are knots in S^3 . Recall from [Gor] that a *ribbon concordance from K_1 to K_0* is a smooth annulus A embedded in $S^3 \times I$ with boundary components $K_0 \times \{0\}$ and $K_1 \times \{1\}$ so that the I -coordinate is a Morse function on A with no local maxima. In this case, we write $K_1 \geq K_0$. In particular, K is ribbon if and only if $K \geq \text{unknot}$. Agol recently proved that $\geq$ gives a partial order on isotopy classes of knots [Agol]. Our search for ribbon disks produced 1,835,130 ribbon concordances $K_1 \geq K_0$; in 1,420,528 cases, K_0 was the unknot. In Section 7, we describe how these give evidence for conjectures that some invariants, such as the hyperbolic volume, are decreasing with respect to this partial order. A sampling of the ribbon concordances we found is shown in Figure 5.

1.14 Category alignment. We now turn to two situations where there seems to be no difference between topological and smooth sliceness. First, recall Lisca showed that a 2-bridge knot is ribbon if and only if it is smoothly slice [Lis]. Moreover, it is posited that a 2-bridge knot K is smoothly slice if and only if it is topologically slice [FM, Conjecture 1]. While there are 2-bridge knots whose topological and smooth slice genera differ [FM], our Theorem 1.7 above still compels us to propose:

1.15 Conjecture. *A prime alternating knot is smoothly slice if and only if it is topologically slice.*

Second, recall that a knot K is *fibred* when the exterior $E_K = S^3 \setminus \nu(K)$ fibers over the circle; here, the fiber surface is necessarily the unique minimum genus Seifert surface. Using knot Floer homology computed via [Sza], we found that 29.9% of the knots in $\mathcal{PS}_{19}$ are fibred. Per Table 5 below, we resolved both topological and smooth sliceness for all but 25 of the fibred knots in $\mathcal{PS}_{19}$. Thus, we were vastly more successful with fibred knots than nonfibred ones (factors of ≈ 200 and ≈ 25 in the smooth and topological settings respectively). This is despite not employing any tests specific to the fibred case, though some exist at least in theory [CG1, CL]. Regardless, in all but the 25 unknown cases, the topological and smooth sliceness matched for fibred knots, suggesting:

1.16 Conjecture. *A fibred hyperbolic knot is smoothly slice if and only if it is topologically slice.*

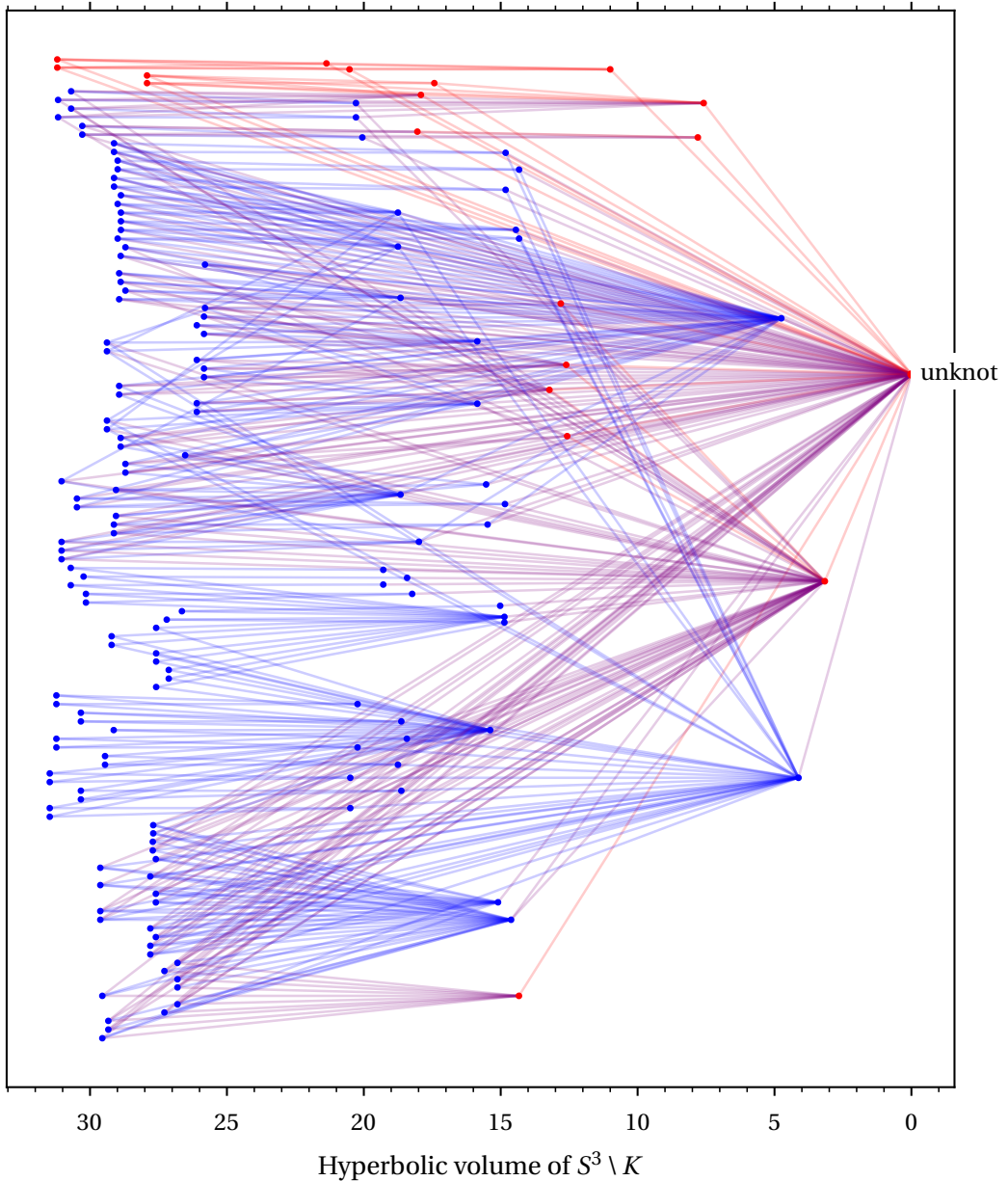


Figure 5. A visualization of ribbon concordances between 168 smoothly slice knots. Each knot is indicated by a dot, where alternating knots are red and other knots are blue. Each edge represents a ribbon concordance $K_1 \geq K_0$ where K_1 always corresponds to the dot farther to the left; that is, the hyperbolic volume of $S^3 \setminus K$ respects the ribbon partial order. The color of each edge is the average of its vertices. Note there are no $K_1 \geq K_0$ where K_1 is alternating and K_0 is not. See Section 7 and Figure 21 for more details.

Here, we restrict to hyperbolic knots because there is only a single non-hyperbolic fibered knot in $\mathcal{PS}_{19}$, namely $17n_{s_{29}}$. Note that a fibered knot that is topologically but not smoothly slice would give rise to a counterexample either to the smooth 4D Poincaré conjecture or to the topological slice-ribbon conjecture discussed in the next subsection [Rub].

1.17 The slice-ribbon conjecture. To search for counterexamples to the slice-ribbon conjecture, one needs methods for showing that a knot K is smoothly slice without exhibiting a ribbon disk. Recall from e.g. [Tei, Lemma 2.5] that a knot K is smoothly slice if and only if there is a ribbon knot J so that $K\#J$ is ribbon. As recounted in Section 2.6, we tried this strategy for several small J . Taking a hint from how [HKL] found a slice disk for $K12a_{990}$, we can sometimes choose J so that we know in advance that $K\#J$ is ribbon concordant to a knot that is simpler than K . Using this method, we discovered more than 500 smoothly slice knots where repeated passes of our ribbon disk search had come up empty. However, we eventually found ribbon disks for all of them; see Section 2.6 for details. Thus we have identified no smoothly slice knots that are not known to be ribbon.

However, the knot $K = 18nh_{00000601}$ from Theorem 1.12 is intriguing in the context of the slice-ribbon conjecture. Unless it and its 0-friend give a counterexample to the smooth 4D Poincaré conjecture, the knot K must be smoothly slice. However, we tried very hard to find a ribbon disk for K to no avail. This makes K a plausible candidate for a knot that is smoothly slice but not ribbon.

A version of the slice-ribbon conjecture in the topological category is as follows. Recall that a knot K is *topologically homotopy ribbon* when it bounds a locally flat disk $F \subset D^4$ so that the inclusion $S^3 \setminus K \hookrightarrow D^4 \setminus F$ is surjective on fundamental groups. (Any ribbon knot is topologically homotopy ribbon [Gor, Lemma 3.1].) The topological slice-ribbon conjecture is that every topologically slice knot is topologically homotopy ribbon. Of the topologically slice knots in Theorem 1.9, all but possibly $18nh_{00000601}$ are in fact topologically homotopy ribbon. This is because knots with $\Delta_K = 1$ are topologically homotopy ribbon by [FQ, Theorem 11.7B] and this is the basis for our primary method (5) above for showing that a knot is topologically slice.

1.18 Inscrutable knots. We next highlight some of the simpler knots not covered by Theorem 1.1 so that they can serve as challenges and benchmarks for future work in this area. Table 1 lists the knots of small crossing number whose smooth sliceness remains unknown; Table 2 is the topological analogue. Table 3 focuses on unknown knots with small hyperbolic volume and Table 4 lists the non-hyperbolic knots; together, these two tables includes all unknown knots in $\mathcal{PS}_{19}$ with Seifert genus 1. Finally, Table 5 lists all unknown knots in $\mathcal{PS}_{19}$ that are fibered.

$K13n65$	$K14n10011$	$K15n17501$	$K15n110439$	$K15n132965$
$K13n3871$	$K14n11063$	$K15n21905$	$K15n117232$	$K15n137921^*$
$K13n3872$	$K14n18909$	$K15n27582$	$K15n120055$	$K15n138033$
$K13n3897$	$K14n18911$	$K15n33471^*$	$K15n121598$	$K15n138051$
$K13n3936$	$K14n21673$	$K15n40402$	$K15n121834$	$K15n140327$
$K13n4582$	$K15n2810$	$K15n49735$	$K15n121916$	$K15n140449$
$K14n3713^*$	$K15n4646$	$K15n53931^*$	$K15n122603$	$K15n144034$
$K14n4425$	$K15n11287$	$K15n94339^*$	$K15n123414$	$K15n146982$
$K14n4621^*$	$K15n11568$	$K15n95989$	$K15n124496$	$K15n154389$
$K14n5486$	$K15n11570$	$K15n95995$	$K15n124640$	$K15n155464$
$K14n9023$	$K15n16056$	$K15n103703$	$K15n130504$	$K15n157903^*$

Table 1. Above are the 55 knots with at most 15 crossings where we could not determine whether they are smoothly slice. With the exception of the seven marked with a “*”, these knots have $\Delta_K = 1$ and hence are topologically slice [FQ, §11.7]. We do not know whether any of the ones marked with “*” are topologically slice; they all have $\Delta_K = (t^2 - t + 1)^2$.

$K14n3713$	$K15n115646^*$	$16n254312$	$16n658612^*$	$16n889116$
$K14n4621$	$K15n137921$	$16n288832$	$16n679074$	$16n898244$
$K15n33471$	$K15n157903$	$16n312764^*$	$16n716954$	$16n911851$
$K15n53931$	$16n86850$	$16n357101$	$16n740455$	$16n914151$
$K15n73236^*$	$16n129260$	$16n374204$	$16n800378^*$	$16n977108$
$K15n77799^*$	$16n178564$	$16n429842$	$16n829217$	$16n982575$
$K15n94339$	$16n180683$	$16n478839^*$	$16n843878$	
$K15n99019^*$	$16n218932^*$	$16n481747$	$16n889114$	

Table 2. Above are the 38 knots with at most 16 crossings where we could not determine whether they are topologically slice. The 9 known not to be smoothly slice are marked with a “*”. Some 30 of these knots have $\Delta_K = (t^2 - t + 1)^2$.

knot	vol	$g_3(K)$	sm	top	knot	vol	$g_3(K)$	sm	top
$19nh_{000000055}$	8.66	2			$19nh_{000003713}$	14.04	6		y
$19nh_{000000156}$	9.96	1			$19nh_{000003849}$	14.09	1		y
$17nh_{0000497}$	11.85	2			$19nh_{000004248}$	14.22	2		
$18nh_{00000597}$	11.92	3		y	$19nh_{000004269}$	14.23	2		y
$18nh_{00000601}$	11.93	5		y	$18nh_{00003640}$	14.36	3		y
$18nh_{00000707}$	12.20	1	n		$16n429842$	14.45	2		
$18nh_{00000752}$	12.30	2			$18nh_{00004014}$	14.49	2		y
$18nh_{00000957}$	12.63	2			$18nh_{00004122}$	14.53	2	n	
$K15n77799$	13.15	2	n		$19nh_{000005728}$	14.60	2	n	
$18nh_{00001632}$	13.31	1	n		$19nh_{000006042}$	14.67	2		
$18nh_{00001818}$	13.44	2			$17nh_{0004389}$	14.73	3		y
$16n481747$	13.59	2			$19nh_{000007003}$	14.85	3		y
$18nh_{00002591}$	13.90	2		y	$18nh_{00005355}$	14.87	3		y
$18nh_{00002656}$	13.93	2		y	$17nh_{0005136}$	14.94	3		y

Table 3. The hyperbolic knots in $\mathcal{PS}_{19}$ of smallest volume whose slice status is not completely known. Here, an “n” in the “sm” column means the knot is not smoothly slice, and a “y” or “n” in the “top” column indicates a known topological slice status. All the unknown knots with $g_3(K) = 1$ appear either in this table or Table 4.

knot	structure	$g_3(K)$	smooth	top
$K15n115646$	Trefoil[3/2]	1	n	
$16n800378$	Trefoil[−1/4]	1	n	
$17ns_{29}$	Fig8[1]	2	n	
$18ns_{51}$	Trefoil[A(1/2, 4/3)]	2	n	
$18ns_{52}$	Trefoil[A(1/2, 3/5)]	2	n	
$18ns_{86}$	Fig8[−1/2]	1		y
$19ns_{006}$	Trefoil[11/6]	1	n	
$19ns_{018}$	Trefoil[15/4]	1	n	
$19ns_{101}$	Trefoil[A(−1/2, 5/7)]	2	n	
$19ns_{157}$	Trefoil[C(−1/2, 1/3, −1/2)]	2	n	
$19ns_{193}$	Trefoil[D(2, −3, −1/2)]	2	n	
$19ns_{244}$	Fig8[−1/3]	2		

Table 4. The 12 non-hyperbolic knots in $\mathcal{PS}_{19}$ whose slice status is not completely known. These are all satellite knots; see [Bur, §5.9] for how to read the structure column. Here an “n” in the “smooth” column means the knot is not smoothly slice, and a “y” in the “top” column means the knot is topologically slice. There are only 21 non-hyperbolic knots in $\mathcal{PS}_{19}$, so we had a much lower success rate in this setting. All the unknown knots with $g_3(K) = 1$ appear either in this table or Table 3.

knot	$g_3(K)$	smooth	top	knot	$g_3(K)$	smooth	top
$17nh_{4549325}$	6			$19nh_{000458099}$	7		
$17ns_{29}$	2	n		$19nh_{000616024}$	4		
$18nh_{00000601}$	5		y	$19nh_{000889975}$	4		
$18nh_{00068958}$	4			$19nh_{000941086}$	4		
$18nh_{00251375}$	4			$19nh_{003772023}$	4		
$18nh_{00446655}$	4			$19nh_{009668884}$	4		
$18nh_{00584034}$	4			$19nh_{010449461}$	4		
$18nh_{01814106}$	4			$19nh_{019331276}$	4		
$18nh_{01872227}$	4			$19nh_{019761670}$	4		
$18nh_{04892103}$	4			$19nh_{029859281}$	5		
$18nh_{10190577}$	4			$19nh_{032455745}$	4		
$19nh_{000054314}$	4			$19nh_{080376121}$	4		
$19nh_{000107587}$	4						

Table 5. The 25 *fibred* knots in $\mathcal{PS}_{19}$ whose slice status is not completely known. Here, the knot $17ns_{29}$ is not smoothly slice by Theorem 6.2, and $18nh_{00000601}$ is topologically slice by Theorem 5.14. The genus of the fibred Seifert surface is listed for each knot.

1.19 Code and data. Of necessity, Theorem 1.9 is the result of a large-scale computer computation. The main software tools we used are SnapPy [CDGW], KnotJob [Sch2], the HFK Calculator [Sza], and SageMath [Sage]. All data and code needed to check our results are permanently archived at [DG], and key parts have been incorporated into the development version of SnapPy. The latter will be part of SnapPy 3.3 but can be used now following the instructions in [DG]. We include in [DG] a quick-access list of PD codes for all 286 knots in $\mathcal{PS}_{19}$ mentioned in this paper by name.

Producing Theorem 1.9 consumed more than 100 CPU-years of computational time over the course of five calendar-years. However, it would take a tiny fraction of that to confirm our results as, for example, we saved a description of each ribbon disk found (see Section 2.10), and these can be quickly checked as valid. Similarly, the data includes the (m, p) parameters of each successful HKL test.

One method we used to try to ensure the correctness of the code is worth mentioning; for concreteness, we describe it in the context of slice obstructions. We initially ran each new slice obstruction on the whole of $\mathcal{PS}_{19}$, even those knots which had already been identified as ribbon. Recall $\mathcal{PS}_{19}$ is more than 42% ribbon knots, and we identified most of those early in our computations. If the code had a bug causing it to randomly report it had found this slice obstruction for 1 in 100,000 knots, we would have seen about 16 ribbon knots being flagged as “not smoothly slice”. Even near the end, when we were attacking the last 50,000 knots with very

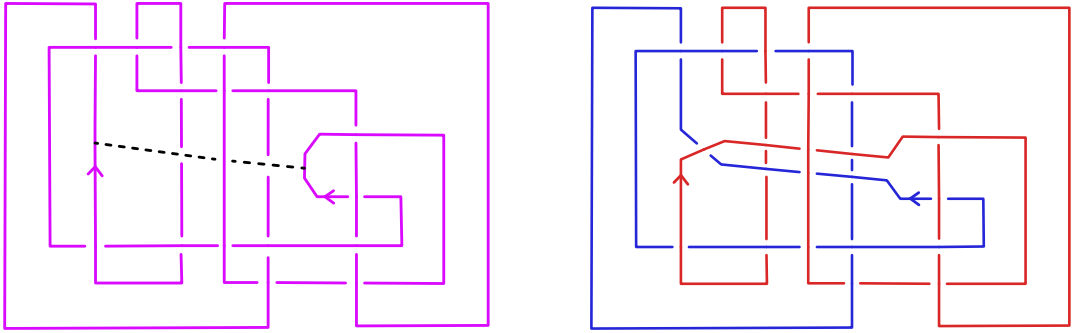


Figure 6. A band move on the knot $K = 16n424914$ at left resulting in the link at right. The band is added along the arc shown by the dotted line and has a half-twist so that the new link L has two components. In fact, L is the unlink, showing that K is ribbon.

time-consuming computations, each run included a substantial portion of knots known not to have the property being checked for as a trap for bugs.

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2 Searching for ribbon disks and concordances

This section is devoted to describing the process we used to show:

2.1 Theorem. *At least 1,633,786 knots in $\mathcal{PS}_{19}$ are ribbon and hence smoothly slice.*

First, we recall a standard description for ribbon disks that we use throughout. For a link L_0 , a *band move* is a 1-handle attachment resulting in a link L_1 with one more component; see Figure 6. A band move is determined by an arc with endpoints in L_0 together with how the band twists around it. A band move encodes an embedded surface in $S^3 \times I$ whose interesting component is a pair of pants P with waist attached to L_0 and cuffs attached to L_1 . Here, one arranges P to have a single critical point with respect to the I -coordinate, which has index 1. When a knot K has a sequence of k band moves resulting in an unlink U , it is ribbon: the cobordism in $S^3 \times I$ made up of k pants combines with disks capping off the components of U in D^4 to form a

length	# bands	% bands	half-twists	# bands	% bands
2	1,684,652	84.6	0	1,079,431	54.2
3	255,129	12.8	1	909,006	45.7
4	43,474	2.2	2	2424	0.12
5	7490	0.4	3	0	0
6	100	0.005			
7	16	0.0008			

Table 6. Statistics on the 1.99 million bands used in Section 2. Here, the length of a band is the number of times it interacts with the diagram, including its endpoints; the band used for Figure 7(d) has length 5. Bands with 3 half-twists were initially included in our search, but since nothing came of them, we subsequently restricted to at most 1 or 2.

ribbon disk. Moreover, any ribbon disk can be described in this way. (The reason a band move is required to increase the number of components is that otherwise the corresponding cobordism results in a non-orientable or otherwise non-planar surface.)

2.2 Diagrammatic bands. For a knot K , there are infinitely-many isotopy classes of arcs that can be used for a band move, to say nothing of the choice of the twist. No bounds are known to exist for the complexity of the arcs needed to encode a ribbon disk, but here we consider the following finite set of possibilities. Given a diagram D for a knot K , we define a *diagrammatic band* as follows. Consider the dual graph to D , which has one vertex for every complementary region of the planar projection of D and one edge for each segment of D between two crossings. Modify this dual graph to create G_D by adding a vertex in the middle of each dual edge where it meets the original D , as in Figure 7(a). A diagrammatic band is one built from a simple path in G_D whose endpoints are on K , together with the over-under information and the number of half-twists; see Figure 7(b, c, d) for an example. For a given path, the parity of the number of half-twists is determined by the requirement that the band increases the number of components. We used diagrammatic bands because of technical challenges to considering bands that cross out of and then back into the same complementary region.

2.3 Search strategy. Our search for ribbon disks was done in several stages. We started with adding only a single band that moreover came from a path in G_D that is minimal length among those joining its endpoints. In Figure 7(b), the shorter arc is minimal length but the longer one used in Figure 7(c) is not. For another example, Figure 6 depicts a band added along a shortest path. Initially, we limited ourselves to

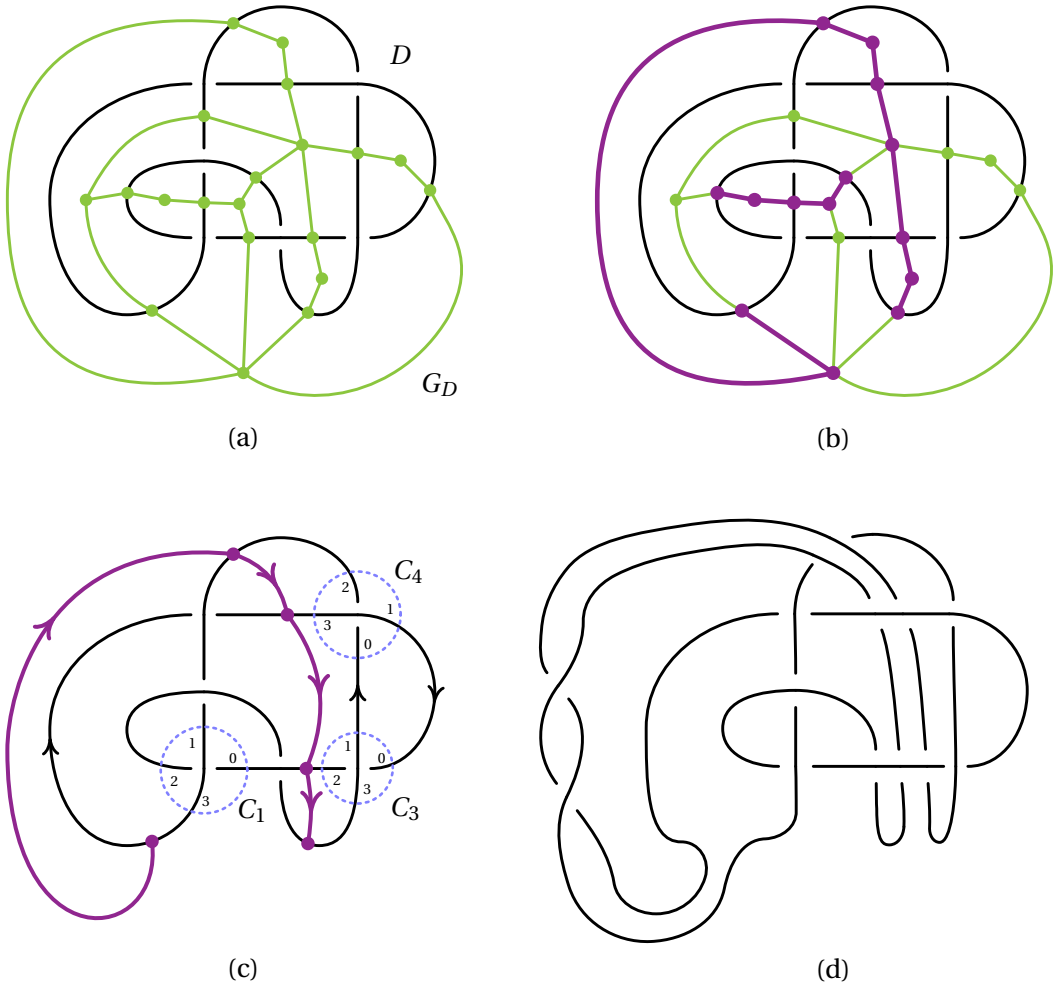


Figure 7. Part (a) shows a diagram D of the knot $K6a3$ with the subdivided dual graph G_D . Part (b) shows two simple paths in G_D that can be used to build diagrammatic bands. Part (c) shows the result of doing one possible band move starting with the path shown in (c). We can record the (oriented) path in (c) by the crossing input pairs to its left, that is $[(C_1, 3), (C_4, 2), (C_4, 3), (C_3, 2), (C_3, 3)]$, and add the over-under information and the number of half-twists (+2 in this case) to specify the band move in (d). In [DG, CDGW], this is `Band([(1, 3), (4, 2), (4, 3), (3, 2), (3, 3)], [True, False, False], 2)`.

three half-twists.

When no shortest path yielded a band giving the unlink, we checked all simple paths in G_D up to a certain length (usually 6). If that failed to find a ribbon disk, we initiated a multi-band search, starting with two bands and working up to four bands.

For the multi-band search, we first considered all links resulting from D by a single band move of the specified type. We pruned this collection of 2-component links by keeping only those where the following necessary conditions for topological sliceness held:

- (1) the linking number between every pair of components was zero,
- (2) the link signature was zero, and
- (3) it satisfied the multivariate Fox-Milnor test; that is, the Alexander nullity was one less than the number of components of the link, and the first non-vanishing Alexander polynomial satisfied the Fox-Milnor condition of [Kaw, Corollary 12.3.14].

For links whose exteriors were hyperbolic, we also removed duplicate copies of the same link from the collection. After pruning, for each link L_i we added a second band in various ways to create a pile of 3-component links. These links were collectively pruned by (1, 2, 3) and consolidated before possibly adding a third band, and so on. Table 6 gives data about the lengths and twists of the bands used.

As part of this process, we also searched for ribbon concordances between knots in $\mathcal{PS}_{19}$ to generate data for Section 7. Suppose a band move on a knot K results in a split two-component link where one component is an unknot and the other component is a knot K' different from K . Such a band then prescribes a ribbon concordance from K to K' with one saddle point and one minimum.

Throughout, identifying a link L as the unlink was primarily done by building an ideal triangulation T of $S^3 \setminus L$, simplifying it to T' using Pachner moves and standard heuristics, computing a presentation for $\pi_1(T')$ and simplifying that, and then finally checking if said presentation had no relators. At least with SnapPy [CDGW], we found this method both faster than using diagrammatic simplification and having an even higher success rate. For example, it correctly identifies the 382 “really hard unknot diagrams” of [ABDE⁺, Appendix A].

2.4 Shaking diagrams. Since we only used diagrammatic bands, the choice of starting diagram is important. We first tried the minimal diagram specified for each knot in [Bur]. When that failed, we generated a diverse collection of possibly slightly larger diagrams and used these to search for bands. Initially, we tried generating other diagrams using random Reidemeister moves, but found the following method

suggested by Question 4 in [DOR, §12] to be more effective. Starting with a diagram D , we computed a triangulation T of its exterior and then used the method of [DOR] to get another diagram D' for the knot. Such “diagram shaking” was used for less than 0.25% of the ribbon knots in Theorem 2.1.

2.5 Caching 2-component ribbon links. For a collection of more than 146,000 ribbon knots in $\mathcal{PS}_{19}$ where we needed at least two bands, we saved the 2-component link created by the first band move. This resulted in a database of 12,143 ribbon links included with [DG]. Two of these links appeared about 19,000 times each ($L10n36 = 10^2_{104}$ and $L10n32 = 10^2_{26}$) and more than 130 appeared at least 100 times, but about 7200 appeared exactly once. With this in hand, it is possible to accelerate the search for ribbon disks, using SnapPy to check if the result of adding a single band is a known ribbon link. Moreover, certain of these links have hard to find ribbon disks. For example, the link R we called `ribbon_2_18_86ba6dfc` initially appeared only for the knot $17nh_{0000980}$. However, we subsequently found 66 other knots where one band move produced R , showing that those knots were ribbon. Overall, this technique identified 206 additional ribbon knots.

2.6 Searching by connected sum with ribbon knots. Another method for detecting smoothly slice knots uses the fact that knot concordance classes form a group under connected sum: for a knot K and a smoothly slice knot J , if $K\#J$ is smoothly slice, then K is as well. Using this, we proceeded as follows: for a knot K , take the connected sum of K with a known ribbon knot R , and then attempt to show that $K\#R$ is ribbon using Section 2.3. (A knot K is smoothly slice if and only if there is a such ribbon knot R [Tei, Lemma 2.5], so in theory this is a universal test.)

We did this search with two specific ribbon knots: the connected sum of a trefoil $T_{2,3}$ with its mirror $T_{2,-3}$, and the connected sum of two copies of the figure-8 knot $K4a1$. The major advantage of using a ribbon knot of the form $A\#B$ is that its connected sum with K can be viewed as adding A and B separately in different places along K , ideally where there are then obvious band moves that decrease the number of crossings. The motivating example comes from [HKL, §8], where $K12a990$ is shown to be smoothly slice using $T_{2,3}\#T_{2,-3}$.

Fix an orientation on a knot K and suppose its diagram has a bigon region whose two sides have parallel orientations. If we form the connected sum of K with a trefoil knot with oppositely signed crossings near this bigon, then we can do a band move that simplifies the resulting 2-component link, as in Figure 8. If we repeat this process with a mirror image of the trefoil next to a different bigon with crossings of opposite sign, this amounts to taking the connected sum with $T_{2,3}\#T_{2,-3}$ and then adding two bands. This results in a diagram for a 3-component link L with two fewer crossings than that of K . We then searched for ribbon disks for L as in Section 2.3. Figure 10

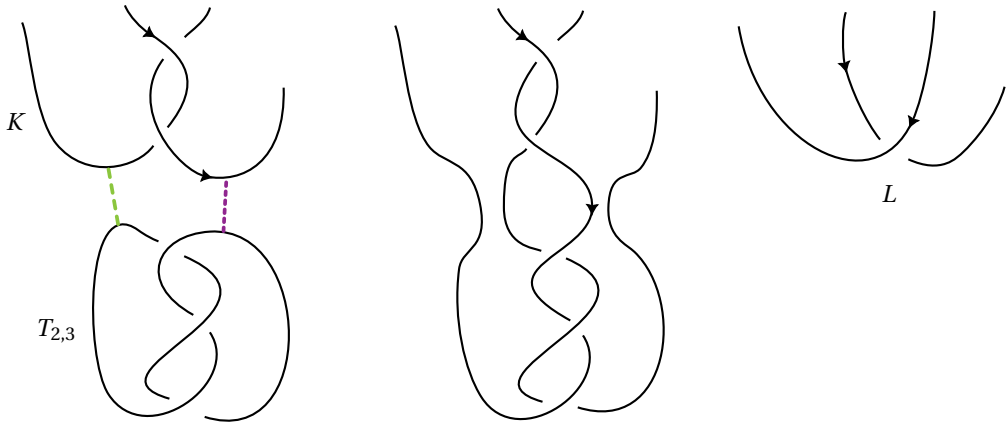


Figure 8. We perform the connected sum of the oriented knot K at left with $T_{2,3}$ along the dark dotted line, and then do an untwisted band move along the light dashed line. The result is shown in the middle. After simplifying, we get the 2-component link L at right which has one fewer crossing than K .

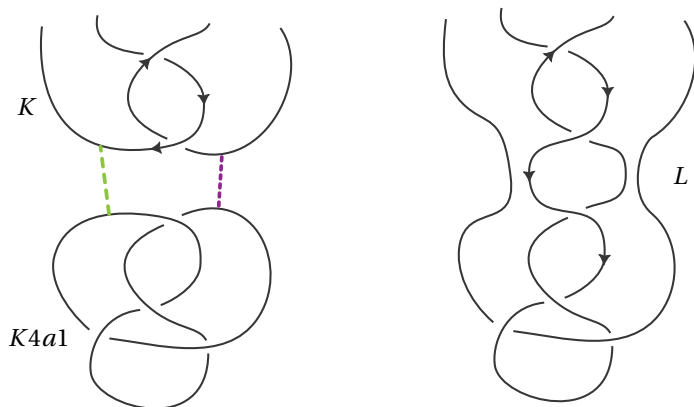


Figure 9. Performing the connected sum of K with $K4a1$ along the dark dotted line and then adding an untwisted band along the light dashed line results in the 2-component link L at left.

shows an example where this strategy works.

The other knot we added was $K4a1\#K4a1$, which is ribbon since $K4a1$ is its own mirror image. As with the trefoils, each $K4a1$ was added along with a band, as in Figure 9. To ensure this is a valid band move, the edges of the bigon must be oriented anti-parallel, as in the figure.

Initially, we found 446 new slice knots in $\mathcal{PS}_{19}$ by adding $T_{2,3}\#T_{2,-3}$ and 67 slice knots by adding $K4a1\#K4a1$. However, by searching harder for ribbon disks, we eventually found them for all 513 of these knots, as discussed in the next subsection.

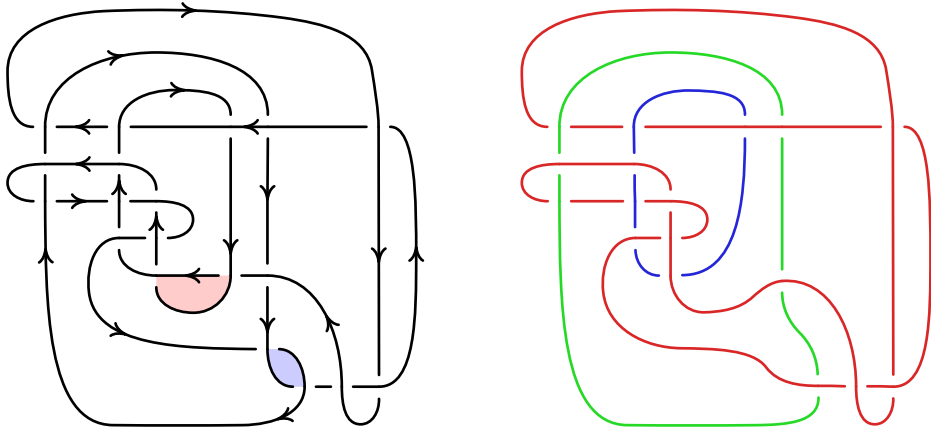


Figure 10. For the knot $K = 19nh_{051162051}$ at left, consider the two shaded bigons, one with positive and one with negative crossings. Taking the connected sum with $T_{2,3} \# T_{2,-3}$ and adding two bands as in Figure 8 and its mirror image yields the 3-component link L at right. As L is an unlink, K is smoothly slice.

2.7 Obscure ribbon disks. For certain knots in $\mathcal{PS}_{19}$, we had reason to suspect that they were ribbon, even though our initial search did not turn up any ribbon disks. This includes the 513 smoothly slice knots from the previous subsection, as well as 41 knots that are 0-friends with a ribbon knot by the method of Section 5.

For these knots, we expanded the parameters of our search (the number of distinct diagrams tried, the number and length of bands checked, etc). In all but one case, namely $18nh_{00000601}$ from Theorem 1.12, we eventually found a ribbon disk. We devoted more computational resources to these suspicious knots than we could afford to do for all the unknown knots in Theorem 1.9. Based on the success rate of the expanded search as compared to that of the original search, we expect there are hundreds of other similarly obscure ribbon disks among the 11,383 knots in $\mathcal{PS}_{19}$ whose smooth slice status is unknown.

The knot $19nh_{051162051}$ of Figure 10 is one example where we initially discovered it was slice by Section 2.6 but eventually found a ribbon disk. Another way we could show that knots were smoothly slice (without a priori knowing that they are ribbon) was by showing that their knot trace was diffeomorphic to that of a ribbon knot (see Section 5.6). Some knots where we initially used the shared knot trace method but later found a ribbon disk are $17nh_{0016322}$, $17nh_{0026540}$, and $17nh_{0298397}$. The knot $17nh_{0016322}$ is depicted in Figure 11(a).

For Section 5, we also identified knots we suspected were smooth slice because they are 0-friends with a ribbon knot (this is a priori weaker than having the same knot trace). For these knots, we also eventually found ribbon disks for all but $18nh_{00000601}$. In particular, we found ribbon disks for $19nh_{000077044}$, $19nh_{000187109}$,

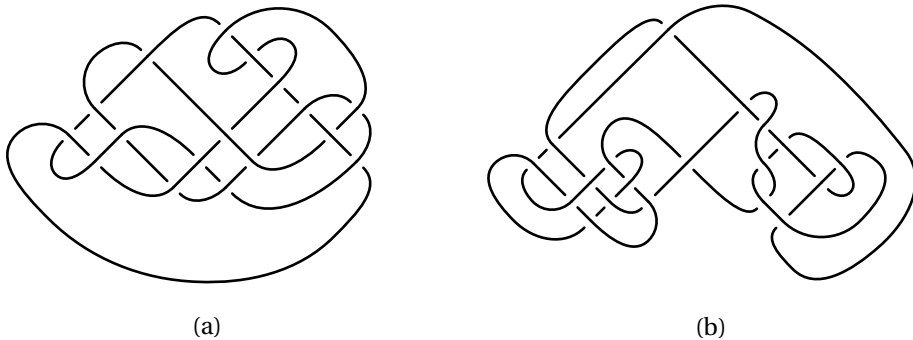


Figure 11. The knot $17nh_{0016322}$ in (a) was initially shown to be smoothly slice by finding a ribbon knot with a diffeomorphic knot trace. We suspected that the knot $19nh_{000077044}$ in (b) was smoothly slice because it shared a 0-surgery with a ribbon knot. For both knots, we eventually found a ribbon disk using an expanded band search.

# bands	# knots	%
1	1,249,604	76.5
2	381,869	23.4
3	2238	0.14
4	75	0.0045

Table 7. The number of bands in the ribbon disks we used in Theorem 2.1. It is an upper bound on the fusion number for each knot.

and $19nh_{003361975}$. The knot $19nh_{000077044}$ is depicted in Figure 11(b).

2.8 Fusion number. The *fusion number* of a ribbon knot is the minimal number of 1-handles needed in a ribbon disk, or equivalently the number of band moves needed to produce the unlink. In our search, we looked for ribbon disks needing up to four bands; the vast majority of the ribbon disks we found only needed one band, with the vast majority of the remainder only needing 2 bands, and so on; see Table 7. Only 75 of the ribbon disks we found used four bands, so we did not search further than that.

There has been much research exploring bounds on the fusion number by computable knot invariants, including the torsion order of knot Floer homology [JMZ], and the X -torsion order in Khovanov homology [Sar, Sch3]. Using KnotJob [Sch2], we computed the X -torsion order for all the knots in $\mathcal{PS}_{19}$, for the fields $\mathbb{Q}$, $\mathbb{F}_3$, and $\mathbb{F}_5$. For our ribbon knots, 582 had X -torsion order two, and the rest had X -torsion order one. The ribbon knots with X -torsion order two must have fusion number at least

two. We found 2-band ribbon disks for all but one of them, namely $18nh_{00098373}$, and hence the rest have fusion number exactly two.

2.9 Related work. As mentioned in Section 1.4, Owens and Swenton studied the 1.2 billion nontrivial prime alternating knots with at most 21 crossings, finding some 664,633 ribbon disks [OS]. They heavily exploited that when such a knot is smoothly slice, it has a “bifactorizable diagram”, a notion that is defined in terms of Goeritz matrices. They then focused on “algorithmic bands” which are short (length at most 3 in our language with at most 1 half-twist) and yield another bifactorizable diagram; in hard cases, they allowed slightly longer “escape bands”. For prime alternating knots with at most 19 crossings, in the end we completely determined which are ribbon in Theorem 1.7 using our generic approach; this includes the 86 knots that were unresolved by [OS]. However, we had to really struggle to find ribbon disks for some knots (e.g. $19ah_{29063095}$, $19ah_{40083815}$, $19ah_{02944101}$, and $19ah_{39877460}$) that are straightforward with the techniques of [OS].

The paper [GHMR] uses machine learning techniques to search for ribbon disks. There, the authors used the same class of diagrammatic bands as we do. However, rather than try all such bands up to some fixed complexity, they use reinforcement learning or Bayesian optimization to generate a collection of bands to add. For knots with at most 14 crossings, [GHMR] arrived at the same list of 1705 ribbon knots as in Theorem 1.9.

2.10 Ribbon certificates. For each ribbon knot in Theorem 2.1, we saved a “ribbon certificate” in [DG]. For such a knot K , this is sequence $(D_1, b_1, D_2, b_2, \dots, b_k, D_{k+1})$ where:

- (1) D_1 is a diagram for K .
- (2) Each D_i is a link diagram and b_i is a diagrammatic band for D_i . Here, D_i is recorded by a PD code and b_i is encoded as in the caption to Figure 7.
- (3) D_{i+1} is a diagram for the link resulting from adding b_i to D_i .
- (4) D_{k+1} is the unlink or one of the links from Section 2.5.

To check a certificate, one needs to verify certain links are isotopic. For example, D_1 may not be the standard diagram for K because of shaking as described in Section 2.4. Similarly, D_{i+1} is typically not the diagram initially generated by adding b_i to D_i ; rather it, is a simplified and possibly shaken version of the same. Most of the time these links have hyperbolic exteriors, in which case SnapPy can easily check the needed link equivalences.

3 Topological obstructions

The main topological slice obstruction we used was that of Herald, Kirk, and Livingston [HKL], which uses twisted Alexander polynomials and the theory developed by Kirk and Livingston [KL] to provide a computationally-efficient form of the Casson-Gordon invariants [CG2]. We introduce several refinements of the methods of [HKL] in Sections 3.6, 3.9, 3.11, and 3.20; we refer to these broadly as *HKL tests*. The main result of this section is:

3.1 Theorem. *Of the knots in $\mathcal{PS}_{19}$, at least 2,211,761 (57.1%) are obstructed from being topologically slice by an HKL test.*

The contributions of the individual techniques is given in Section 3.21.

3.2 Basic setup. We first outline how these invariants work, following [HKL, KL, FV]. For a knot K in S^3 , let E_K be its exterior $S^3 \setminus \nu(K)$, and let B_m be the m -fold cyclic cover of S^3 branched over K for some $m \geq 2$. Consider $H_1(B_m; \mathbb{Z})$ and a homomorphism $\chi: H_1(B_m; \mathbb{Z}) \rightarrow \mathbb{Z}/q\mathbb{Z}$ for some integer $q \geq 2$. (We assume here that $H_1(B_m; \mathbb{Z})$ is finite, which can easily be arranged or checked directly.) By identifying $\mathbb{Z}/q\mathbb{Z}$ with the multiplicative group generated by $\zeta_q = e^{2\pi i/q}$ in $\mathbb{Q}(\zeta_q)$, we can see ρ as a representation of $H_1(B_m; \mathbb{Z})$ on $\mathbb{Q}(\zeta_q)$, which induced down from $\pi_1(m\text{-fold cyclic cover of } E_K)$ to $\pi_1(E_K)$, gives a representation $\rho_\chi: \pi_1(E_K) \rightarrow \text{GL}_m(\mathbb{Q}(\zeta_q))$. If Z_K is the 0-framed Dehn surgery on K , then ρ_χ extends to $\pi_1(Z_K)$ and the criterion will be in terms of the twisted Alexander polynomial of (Z_K, ρ_χ) , which is a Laurent polynomial in $\mathbb{Q}(\zeta_q)[t^{\pm 1}]$ that we denote as Δ_χ but is $\Delta_{Z_K, (\mathbb{Q}[\zeta_q[t^{\pm 1}]]^m}$ in [HKL], $\Delta_1(Z_K; \rho_\chi)$ in [KL], and $\Delta_{Z_K}^{\alpha(m, \chi)}$ in [FV].

The basic idea is that if K is topologically slice, then B_m is the boundary of the 4-manifold N_m obtained by taking the m -fold cyclic cover of D^4 branched over the slice disk. If χ extends to $H_1(N_m; \mathbb{Z})$, possibly after enlarging the range to $\mathbb{Z}/q^k\mathbb{Z}$, then $\Delta_\chi(t)$ must be a *norm*, that is, equal to $f(t)\bar{f}(t)$ up to units in $\mathbb{Q}(\zeta_q)[t^{\pm 1}]$; here $\bar{f}$ is the result of applying the involution to $\mathbb{Q}(\zeta_q)[t^{\pm 1}]$ that interchanges t and t^{-1} and is complex conjugation on $\mathbb{Q}(\zeta_q)$. To understand when χ might extend to N_m , one induces the notion of an *invariant metabolizer*, a subgroup A of $H := H_1(B_m; \mathbb{Z})$ where A is invariant under the $\mathbb{Z}/m\mathbb{Z}$ -action on H , the orders satisfy $|A|^2 = |H|$, and A is its own orthogonal complement under the nonsingular linking form $lk: H \otimes H \rightarrow \mathbb{Q}/\mathbb{Z}$. This is relevant because the kernel of $H \rightarrow H_1(N_m; \mathbb{Z})$ is necessarily such an invariant metabolizer. The following is a reformulation of the results of Section 6 of [KL] as in [FV]:

3.3 Theorem [KL]. Suppose m and q are prime powers where q is odd. If K is topologically slice, then there exists an invariant metabolizer $A \subset H$ such that every $\chi: H \rightarrow \mathbb{Z}/q\mathbb{Z}$ with $A \subset \ker(\chi)$ gives a Δ_χ that is a norm.

To use this as a slice obstruction, one considers all possible invariant metabolizers A and for each one finds at least one χ for which Δ_χ is not a norm; the precise tests we used are discussed in Section 3.4. As in [HKL], we do not actually compute the form lk , and so work with the possibly larger class of A that are invariant and have the correct size.

3.4 The original HKL test. Until Section 3.20, we will only consider the special case of Theorem 3.3 when q is prime. We also assume that $\gcd(m, q) = 1$ as otherwise it turns out that $H^1(B_m; \mathbb{F}_q)$ will be zero. Since q is prime, any $\chi: H \rightarrow \mathbb{Z}/q\mathbb{Z}$ can be viewed as an element of $H^1(B_m; \mathbb{F}_q)$. Since m and q are coprime, the polynomial $x^m - 1$ in $\mathbb{F}_q[x]$ is separable, and, by the theory of rational canonical form for matrices [DF, §12.2], the irreducible representations of $\mathbb{Z}/m\mathbb{Z}$ on finite-dimensional $\mathbb{F}_q$ -vector spaces correspond to the irreducible factors of $x^m - 1$ in $\mathbb{F}_q[x]$. Indeed, given an irreducible representation ψ of $\mathbb{Z}/m\mathbb{Z}$ on such a vector space V , the corresponding factor of $x^m - 1$ is the characteristic polynomial of the ψ -action of the preferred generator of $\mathbb{Z}/m\mathbb{Z}$.

Switching to the language of modules over $\mathbb{F}_q[\mathbb{Z}/m\mathbb{Z}]$, decompose $\bar{H} = H_1(B_m; \mathbb{F}_q)$ into irreducible submodules as

$$\bar{H} = \bigoplus_{i=1}^k V_i^{e_i} \quad \text{where the } V_i \text{ are distinct and all } e_i \geq 1.$$

Let $\bar{A}$ be the image of a metabolizer $A \subset H_1(B_m; \mathbb{Z})$ in $\bar{H}$; by Lemma 8.2 of [HKL], it is a *proper* subspace of $\bar{H}$. By Schur's lemma, there are $\mathbb{Z}/m\mathbb{Z}$ -invariant subspaces W_i of $V_i^{e_i}$ so that $\bar{A} = \bigoplus_{i=1}^k W_i$.

For an $\mathbb{F}_q$ -vector space W , we denote the dual vector space of linear functionals $W \rightarrow \mathbb{F}_q$ by W^* , and further identify $H^1(B_m; \mathbb{F}_q)$ with $\bar{H}^*$. When all the multiplicities $e_i = 1$, we have the following principle that underlies the applications in Section 10 of [HKL]:

3.5 Lemma. Suppose $\bar{H}$ decomposes into $\bigoplus_{i=1}^k V_i$ where the V_i are distinct irreducibles. Suppose each V_i^* contains a χ_i where Δ_{χ_i} is not a norm (here χ_i is defined as 0 on the other V_j). Then K is not topologically slice.

Proof. As $\bar{A}$ is a proper subspace of $\bar{H}$ and is the direct sum of a subset of the V_i , after reindexing the V_i we may assume $\bar{A}$ is contained in $\bigoplus_{i=2}^k V_i$. Then $\bar{A}$ is contained in $\ker(\chi_1)$, and, since Δ_{χ_1} is not a norm, Theorem 3.3 implies that K is not topologically slice. $\square$

3.6 A refined HKL test. We now analyze the image $\bar{A}$ of the metabolizer more carefully, leading to Theorem 3.8 which improves Lemma 3.5. As in [HKL, §8], for any prime q we consider the localization $H_{(q)} = H \otimes \mathbb{Z}_{(q)}$ of H , which concretely is the subgroup of elements whose order is a power of q ; hence $H = \bigoplus \{H_{(q)} \mid q \text{ prime}\}$ where each $H_{(q)}$ is $\mathbb{Z}/m\mathbb{Z}$ -invariant. The subgroups $H_{(q)}$ are orthogonal under the linking form lk , and hence the restriction of lk to each $H_{(q)}$ is nondegenerate. Moreover, the metabolizer similarly decomposes into the sum of $A_{(q)} \leq H_{(q)}$, and one can check that $|A_{(q)}|^2 = |H_{(q)}|$. Here is the key fact that strengthens Lemma 8.2 of [HKL] and allows us to go beyond Lemma 3.5:

3.7 Lemma [Saw]. As $\mathbb{F}_q$ -vector spaces, $\dim \bar{A} \leq \frac{1}{2} \dim \bar{H}$. Moreover, $H_{(q)} / A_{(q)} \cong A_{(q)}$.

Proof. For any group N , define $N^\vee = \text{Hom}(N, \mathbb{Q}/\mathbb{Z})$. Set $G = H_{(q)}$. Since lk on G is nondegenerate, the natural map $\iota: G \rightarrow G^\vee$ defined by $g \mapsto lk(g, \cdot)$ is injective. For any $M \leq G$, the surjection $G \twoheadrightarrow G/M$ induces an injection $(G/M)^\vee \hookrightarrow G^\vee$, and we identify $(G/M)^\vee$ with its image. Note that $M^\perp$ is the preimage of $(G/M)^\vee$ under ι . Taking M to be the metabolizer $A_{(q)}$, from $M = M^\perp$ we get that $M \cong (G/M)^\vee$; since $N^\vee \cong N$ for any finite group, we have $M \cong G/M$ proving the second claim.

Recall that $\text{rank } N$ is the minimal number of generators of a group N ; when N is a q -group, $\text{rank } N = \dim_{\mathbb{F}_q} (N \otimes \mathbb{Z}/q\mathbb{Z})$. Let $\bar{G} = G/qG$ and $\bar{M}$ be the image of M in $\bar{G}$. Note that $\text{rank}(G/M) = \dim \bar{G} - \dim \bar{M}$. Since $M \cong G/M$, we have

$$\text{rank } M + \dim \bar{M} = \dim \bar{G}.$$

Since $\dim \bar{M} \leq \text{rank } M$, we have $2 \dim \bar{M} \leq \dim \bar{G}$ as needed. $\square$

3.8 Theorem. Suppose $\bar{H}$ decomposes into distinct irreducibles as $\bigoplus_{i=1}^k V_i^{e_i}$. We say that V_i is weakly obstructed when there is some $\chi \in (V_i^{e_i})^*$ where Δ_χ is not a norm; it is strongly obstructed if for every proper invariant subspace $W \subset V_i^{e_i}$ there is a $\chi \in (V_i^{e_i})^*$ with $W \subset \ker \chi$ where Δ_χ is not a norm. (When $e_i = 1$, these notions coincide.) Define

$$\delta(V_i) = \begin{cases} \dim V_i^{e_i} & \text{if } V_i \text{ is strongly obstructed} \\ \dim V_i & \text{if } V_i \text{ is weakly obstructed} \\ 0 & \text{otherwise} \end{cases}$$

If $\sum_{i=0}^k \delta(V_i) > \frac{1}{2} \dim \bar{H}$, then K is not topologically slice.

Proof. As noted above, $\bar{A}$ is the direct sum of invariant subspaces $A_i \subset V_i^{e_i}$. If $\dim A_i \geq \delta(V_i)$ for all i , then

$$\dim \bar{A} = \sum_{i=0}^k \dim A_i \geq \sum_{i=0}^k \delta(V_i) > \frac{1}{2} \dim \bar{H}$$

contradicting Lemma 3.7. Therefore, we have an i with $0 \leq \dim A_i < \delta(V_i)$. If V_i is strongly obstructed so that $\delta(V_i) = \dim V_i^{e_i}$, this means $A_i \neq V_i^{e_i}$; if V_i is weakly obstructed so $\delta(V_i) = \dim(V_i)$, we get $A_i = \{0\}$. In either case, there is a $\chi \in (V_i^{e_i})^*$ vanishing on all of $\bar{A}$, and Theorem 3.3 applies to show K is not topologically slice. $\square$

3.9 Checking if a polynomial is a norm. The paper [HKL] does not give a general procedure for deciding when Δ_χ is a norm; rather, the authors used various techniques (e.g. a form of Gauss's lemma over $\mathbb{Z}[\zeta_q]$) to show that the Δ_χ appearing in their computations were not norms. Below in Proposition 3.10, we give a general criterion that is easy to implement provided one can factor polynomials with coefficients in a number field, which is a well-studied problem [Coh, Sage].

We work in the setting of a subfield F of $\mathbb{C}$ that is invariant under complex conjugation. Let $R = F[t, t^{-1}]$ be its ring of Laurent polynomials. Consider the involutive ring isomorphism that combines complex conjugation with $t \mapsto 1/t$; that is, if $f(t) = \sum_n a_n t^n$, set $\bar{f}(t) = \sum_n \bar{a}_n t^{-n}$ where the $a_n \in F$ and the sum is actually finite. Recall that $f \in R$ is a norm when there is a $g \in R$ with $f = ug\bar{g}$ for some unit u of R (here $u = at^n$ for some $a \in F$). We have the following test:

3.10 Proposition. Suppose $f \in R$ is $up_1^{e_1}p_2^{e_2}\cdots p_k^{e_k}$ where $u \in R$ is a unit and the $p_i \in R$ are non-associate irreducibles. Then f is a norm if and only if for every i either

- (1) $\bar{p}_i$ is an associate of p_i and e_i is even, or
- (2) $\bar{p}_i$ is an associate of some other p_j and $e_i = e_j$.

Proof. Recall that R is a principal ideal domain (it is a localization of the PID $F[t]$) and hence has unique factorization. To avoid messing with units, we work with ideals rather than elements. If f has the required factorization we can write

$$(f) = (q_1)^{2d_1} \cdots (q_\ell)^{2d_\ell} ((r_1)(\bar{r}_1))^{e_1} \cdots ((r_m)(\bar{r}_m))^{e_m}$$

where each $(q_i) = (\bar{q}_i)$ and all of the (q_i) , (r_i) , and $(\bar{r}_i)$ are distinct. We then take $(g) = (q_1)^{d_1} \cdots (q_\ell)^{d_\ell} (r_1)^{e_1} \cdots (r_m)^{e_m}$ to show (f) is a norm.

Conversely, suppose f is a norm, that is $(f) = (g)(\bar{g})$, and consider a factorization $(g) = (q_1)^{d_1} \cdots (q_k)^{d_k}$ where the (q_i) are distinct prime ideals. As $h \mapsto \bar{h}$ is a ring

17nh ₀₆₃₀₈₈₉	18nh ₀₉₂₉₂₅₁₈	18nh ₁₃₇₉₈₇₀₂	18nh ₂₅₈₇₂₂₀₅
19nh ₀₀₀₁₃₀₅₆₃	19nh ₀₀₀₁₃₀₅₆₄	19nh ₀₀₁₅₆₁₉₄₈	19nh ₀₀₁₇₄₆₁₉₉
19nh ₀₀₁₇₈₅₂₈₇	19nh ₀₁₅₀₈₈₀₅₈	19nh ₀₂₀₇₄₆₁₀₂	19nh ₀₂₆₈₂₄₆₇₁
19nh ₀₃₂₃₉₃₀₇₆	19nh ₀₃₅₃₂₀₂₄₈	19nh ₀₃₅₄₈₇₆₈₂	19nh ₀₅₅₈₆₇₆₄₇
19nh ₀₆₈₈₇₂₁₁₅	19nh ₀₈₃₅₄₇₅₇₀	19nh ₁₀₉₅₉₃₃₇₄	19nh ₁₄₄₁₈₆₂₄₇
19nh ₁₄₄₁₈₆₂₄₈	19nh ₁₄₆₇₈₉₆₈₃	19nh ₁₅₀₀₈₁₂₁₆	19nh ₁₆₉₈₅₂₄₀₅

Table 8. We do not know whether these 24 knots are smoothly or topologically slice, but they are not ribbon, or even topologically homotopy ribbon, by Theorem 3.12 using $m = 3$ and $k = 1$. They all have $\Delta_K = t^4 - 2t^3 + 3t^2 - 2t + 1$.

isomorphism, we have that $(\bar{q}_1)^{d_1} \cdots (\bar{q}_k)^{d_k}$ is such a factorization of $(\bar{g})$. Consider the factorization of (f) gotten by combining those for (g) and $(\bar{g})$ and grouping common factors. If $(\bar{q}_i) = (q_i)$, then (q_i) occurs in (f) to the power $2d_i$ as needed for (1). If $(\bar{q}_i) = (q_k)$ for some $k \neq i$, then (q_i) and $(\bar{q}_i)$ both occur in (f) to the power $d_i + d_k$ as needed for (2). Finally, if $(\bar{q}_i) \neq (q_k)$ for all k , then (q_i) and $(\bar{q}_i)$ both occur in (f) to the power d_i , and so (f) has the required factorization. $\square$

3.11 Allowing $q = 2$. We now explain why Theorem 3.3 excludes $q = 2$ and give a variant that can be applied even in this case. As noted in Section 3.2, one may have to enlarge the range of χ to $\mathbb{Z}/q^k\mathbb{Z}$ to extend it over the 4-manifold N_m bounded by B_m . Thus while Δ_χ is in $\mathbb{Q}(\zeta_q)[t^{\pm 1}]$, if the knot is topologically slice, what we learn from Proposition 6.1 of [KL] is that Δ_χ is a norm in the larger ring $\mathbb{Q}(\zeta_{q^k})[t^{\pm 1}]$ where k is unknown. By Lemma 6.4 of [KL], for odd d the latter condition is equivalent to being a norm in the original $\mathbb{Q}(\zeta_q)[t^{\pm 1}]$, giving Theorem 3.3. However for $q = 2$, we have examples such as $f(t) = t^4 + 4t^2 + 1$ which is irreducible in $\mathbb{Q}[t^{\pm 1}] = \mathbb{Q}(\zeta_2)[t^{\pm 1}]$ and so not a norm there, but $f = g\bar{g}$ for $g = t^2 + (\sqrt{2}i)t + 1$ in $\mathbb{Q}(\zeta_8)[t^{\pm 1}]$.

If the knot K is ribbon, then $H_1(B_m; \mathbb{Z}) \rightarrow H_1(N_m; \mathbb{Z})$ is onto and there is no need to extend χ . In fact, one has:

3.12 Theorem. *Suppose m is an odd prime power. If K is topologically homotopy ribbon, then there exists an invariant metabolizer $A \subset H$ such that every $\chi: H \rightarrow \mathbb{Z}/2^k\mathbb{Z}$ with $A \subset \ker(\chi)$ gives a Δ_χ that is a norm.*

Proof. Recall that K is topologically homotopy ribbon when it bounds a locally flat disk $F \subset D^4$ so that the inclusion $S^3 \setminus K \hookrightarrow D^4 \setminus F$ is surjective on fundamental groups. (Any ribbon knot is topologically homotopy ribbon [Gor, Lemma 3.1].) Consequently, the inclusion $B_m \rightarrow N_m$ gives a surjection $H \rightarrow H_1(N_m; \mathbb{Z})$; the kernel of this is A . As $A \subset \ker(\chi)$, we get that χ extends to $H_1(N_m; \mathbb{Z})$ and hence Δ_χ is a norm by Proposition 6.1 of [KL]. $\square$

3.13 Remark. Theorem 3.12 gives a ribbon obstruction that is not obviously a topological slice obstruction. We found 24 knots in $\mathcal{PS}_{19}$ where Theorem 3.12 applies, but we have been unable to determine whether the knot is smoothly or topologically slice. These are listed in Table 8, and are thus candidates for knots that are slice but not ribbon.

To get a genuine topological slice obstruction when $q = 2$, we need a criterion for showing that $f \in \mathbb{Q}[t^{\pm 1}]$ is not a norm in $\mathbb{Q}(\zeta_{2^k})[t^{\pm 1}]$ for all k ; equivalently, setting $\mathbb{Q}(\zeta_{2^\infty}) = \bigcup_k \mathbb{Q}(\zeta_{2^k})$, we want to know whether f is a norm in $\mathbb{Q}(\zeta_{2^\infty})[t^{\pm 1}]$. Proposition 3.15 below gives the needed test, allowing us to apply Lemma 3.5 and Theorem 3.8 when $q = 2$.

For use in Section 3.20, we consider a slightly more general question. Suppose $K = \mathbb{Q}(\zeta_{2^n})$ and $f \in K[t^{\pm 1}]$ is irreducible. Let $L_f \subset \mathbb{C}$ be the splitting field of f and take $\text{Gal}(f) = \text{Gal}(L_f/K)$. For any finite group G , we set:

$$d(G) = \max \{k \geq 0 \mid \exists H \trianglelefteq G \text{ with quotient } \mathbb{Z}/2^k\mathbb{Z} \text{ or } \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2^{k-1}\mathbb{Z}\} \quad (3.14)$$

and further define $d(f) = d(\text{Gal}(f))$.

Our test is the following, where the case $K = \mathbb{Q}$ corresponds to $n = 1$, not $n = 0$:

3.15 Proposition. *Let $K = \mathbb{Q}(\zeta_{2^n})$ with $n \geq 1$. Suppose $f \in K[t^{\pm 1}]$ factors as $up_1^{e_1}p_2^{e_2}\cdots p_k^{e_k}$ where u is a unit and the p_i are non-associate irreducibles. Then f is a norm in $\mathbb{Q}(\zeta_{2^\infty})[t^{\pm 1}]$ if and only if for every i one of the following holds:*

- (1) $\bar{p}_i$ is an associate of p_i and e_i is even, or
- (2) $\bar{p}_i$ is an associate of p_i with e_i odd, and p_i is a norm in $\mathbb{Q}(\zeta_{2^\ell})[t^{\pm 1}]$ with $\ell = n + d(p_i) + 1$, where $d(p_i)$ is from (3.14), or
- (3) $\bar{p}_i$ is an associate of some other p_j and $e_i = e_j$.

The following will be needed for Proposition 3.15:

3.16 Lemma. *Every finitely-generated subfield L of $\mathbb{Q}(\zeta_{2^\infty})$ is Galois over $\mathbb{Q}$ and has $[L : \mathbb{Q}] = 2^k$ for some $k \geq 0$. Moreover, the field $\mathbb{Q}(\zeta_{2^\infty})$ has exactly three subfields L with $[L : \mathbb{Q}] = 2^k$ for each $k \geq 1$. These are $\mathbb{Q}(\zeta_{2^{k+1}})$, $\mathbb{Q}(\cos(\pi/2^{k+1}))$, and $\mathbb{Q}(\sin(\pi/2^{k+1})i)$. All such L are contained in $\mathbb{Q}(\zeta_{2^{k+2}})$ and have $\text{Gal}(L/\mathbb{Q})$ either $\mathbb{Z}/2^k\mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2^{k-1}\mathbb{Z}$. Finally, if $M \subset \mathbb{Q}(\zeta_{2^\infty})$ is finitely-generated and contains L , then M/L is Galois with group $\mathbb{Z}/2^\ell\mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2^{\ell-1}\mathbb{Z}$ where $[M : L] = 2^\ell$.*

Proof. Any finitely-generated L is contained in some $\mathbb{Q}(\zeta_{2^\ell})$. Recall

$$G = \text{Gal}(\mathbb{Q}(\zeta_{2^\ell})/\mathbb{Q}) \cong (\mathbb{Z}/2^\ell\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2^{\ell-2}\mathbb{Z}$$

and that any subfield L of $\mathbb{Q}(\zeta_{2^\ell})$ corresponds to a subgroup $H \leq G$. As G is abelian, H is normal and so $L/\mathbb{Q}$ is Galois. As $[L : \mathbb{Q}] = |G/H|$ divides $|G| = 2^{\ell-1}$, we have proved the first claim.

Now fix $k \geq 1$. We first analyze the subfields L of $\mathbb{Q}(\zeta_{2^\ell})$ of degree 2^k over $\mathbb{Q}$ when $\ell \geq k+2$. Let $H \leq G$ be the subgroup corresponding to L . Now, G/H is generated by two elements, one of which has order at most two, and so is either $\mathbb{Z}/2^k\mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2^{k-1}\mathbb{Z}$. Up to automorphisms of the target group, you can check that there are two epimorphisms from G onto $\mathbb{Z}/2^k\mathbb{Z}$ and one onto $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2^{k-1}\mathbb{Z}$. Thus $\mathbb{Q}(\zeta_{2^\ell})$ has exactly three subfields of degree 2^k over $\mathbb{Q}$.

For any $m > \ell$, we similarly have $\mathbb{Q}(\zeta_{2^m})$ has three subfields of degree 2^k over $\mathbb{Q}$, and therefore these subfields must be the ones inherited from $\mathbb{Q}(\zeta_{2^\ell}) \subset \mathbb{Q}(\zeta_{2^m})$. Since any finitely-generated subfield of $\mathbb{Q}(\zeta_{2^\infty})$ is contained in some $\mathbb{Q}(\zeta_{2^m})$, we have proven that $\mathbb{Q}(\zeta_{2^\infty})$ has exactly three subfields of degree 2^k , specifically those coming from $M = \mathbb{Q}(\zeta_{2^{k+2}})$. To identify them concretely, first note that each corresponds to the fixed field of one of the three involutions in $\text{Gal}(M/\mathbb{Q})$. If $\zeta = \zeta_{2^{k+2}}$, these are $\zeta \mapsto \bar{\zeta}$, $\zeta \mapsto -\zeta$, and $\zeta \mapsto -\bar{\zeta}$, whose fixed fields are $\mathbb{Q}(\cos(\pi/2^{k+1}) = \zeta + \bar{\zeta})$, $\mathbb{Q}(\zeta_{2^{k+1}} = \zeta^2)$, and $\mathbb{Q}(\sin(\pi/2^{k+1})i = \zeta - \bar{\zeta})$, respectively.

The final claims about M/L now follow from our knowing $\text{Gal}(M/\mathbb{Q})$ is $\mathbb{Z}/2^m\mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2^{m-1}\mathbb{Z}$ and that the only quotients of these groups have the same form. $\square$

3.17 Corollary. *Let $K = \mathbb{Q}(\zeta_{2^n})$ for $n \geq 1$ and suppose $f \in K[t^{\pm 1}]$ is irreducible. If f is a norm over $\mathbb{Q}(\zeta_{2^\infty})$ then it is a norm over $\mathbb{Q}(\zeta_{2^\ell})$ with $\ell = n + d(f) + 1$.*

Proof. Set $R = \mathbb{Q}(\zeta_{2^\infty})[t^{\pm 1}]$. Suppose $f = ug\bar{g}$ with $g \in R$ and $u \in R$ a unit, i.e. $u = at^d$ for nonzero $a \in \mathbb{Q}(\zeta_{2^\infty})$. Multiplying by units in R , we can assume that f and g are genuine polynomials in $\mathbb{Q}(\zeta_{2^\infty})[t]$ with nonzero constant terms. Then $f = at^n g\bar{g}$ where $n = \deg(g)$ and $a \in \mathbb{Q}(\zeta_{2^\infty})$.

Let L_f be the splitting field of f over K . Writing $g(t) = \prod_{i=1}^n (t - \theta_i)$, we have $\theta_i \in L_f$ so $g \in L_f[t^{\pm 1}]$. Now $\bar{g} = \prod_{i=1}^n (1/t - \bar{\theta}_i)$, so each $1/\bar{\theta}_i$ is a root of f and hence $\bar{\theta}_i$ is also in L_f . Thus $\bar{g} \in L_f[t^{\pm 1}]$ as well. Solving for a in $f = at^n g\bar{g}$, we see that $a \in L_f$ as well. Thus $f = at^n g\bar{g}$ shows that f is a norm over the field $M = L_f \cap \mathbb{Q}(\zeta_{2^\infty})$.

Now M is Galois over K as both L_f and $\mathbb{Q}(\zeta_{2^\infty})$ are, and so M corresponds to a normal subgroup H of $\text{Gal}(f)$. By the last claim of Lemma 3.16, $\text{Gal}(M/K)$ is $\mathbb{Z}/2^k\mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2^{k-1}\mathbb{Z}$ for some $k \geq 0$. As $\text{Gal}(M/K)$ is also $\text{Gal}(f)/H$, we have $k \leq d(f)$ by (3.14). So $[M : \mathbb{Q}] = [M : K][K : \mathbb{Q}] = 2^{k+(n-1)} \leq 2^{n+d(f)-1}$. Again by Lemma 3.16, we have $M \subset \mathbb{Q}(\zeta_{2^\ell})$ where $\ell = n + d(f) + 1$, and so f is a norm over $\mathbb{Q}(\zeta_{2^\ell})$ as needed. $\square$

We now prove the main result of this subsection.

Proof of Proposition 3.15. The “only if” direction is straightforward and left to you. For the other, suppose f is a norm in some $\mathbb{Q}(\zeta_{2^\ell})[t^{\pm 1}]$. By Proposition 3.10, as ideals in $\mathbb{Q}(\zeta_{2^\ell})[t^{\pm 1}]$ we have:

$$(f) = (q_1)^{2d_1} \cdots (q_\ell)^{2d_\ell} (r_1)(\bar{r}_1)^{e_1} \cdots (r_m)(\bar{r}_m)^{e_m}$$

where each $(q_i) = (\bar{q}_i)$ and all of the (q_i) , (r_i) , and $(\bar{r}_i)$ are distinct. Consider the action of $G = \text{Gal}(\mathbb{Q}(\zeta_{2^\ell})/K)$ on these ideals, which permutes them amongst themselves. Since the absolute Galois group $\overline{\text{Gal}}(\mathbb{Q}(\zeta_{2^\ell})/\mathbb{Q})$ is abelian and complex conjugation is one of its elements, we have $\sigma \cdot (g) = \sigma \cdot (\bar{g})$ for all $\sigma \in G$ and $g \in \mathbb{Q}(\zeta_{2^\ell})[t^{\pm 1}]$. So the G -orbit of (q_1) consists only of certain other (q_i) ; by uniqueness of factorizations, we have $d_i = d_1$ in this case. The product of the ideals in the G -orbit of (q_1) gives an ideal (p) generated by a $p \in K[t^{\pm 1}]$ which is irreducible there, and indeed p^{2d_1} is a factor of f in case (1). The same is true for the other orbits of the (q_i) .

Turning to (r_1) , there are two cases. If $(\bar{r}_1)$ is in a different G orbit than (r_1) , the same is true of any other (r_j) or $(\bar{r}_j)$ in its orbit. Relabeling, we can assume the G orbit of (r_1) is all (r_i) with $1 \leq i \leq k$; here $e_i = e_1$ for such i . The product of these (r_i) gives (p) with $p \in K[t^{\pm 1}]$ irreducible. Now $(\bar{p}) = (\bar{r}_1)(\bar{r}_2) \cdots (\bar{r}_k)$ is distinct from (p) , so p^{e_1} is a factor of f of in case (3). Finally, consider when $(\bar{r}_1)$ is in the same G orbit as (r_1) . In this case, if (p) is the product of all the ideals in the orbit of (r_1) , we get the usual factor p^{e_1} of f . If e_1 happens to be even, we are in case (1), and otherwise we apply Corollary 3.17 to land in case (2). $\square$

3.18 Remark. Computing the Galois group of a polynomial is a highly nontrivial task. This is especially the case when the base field is not $\mathbb{Q}$, where the only implementation we are aware of is in Magma [BCP]. However, the following result allows us to avoid computing the Galois group in practice. For example, if $f \in \mathbb{Q}[t]$ has $\deg(f) \leq 14$, it shows we need only worry about f being a norm over $\mathbb{Q}(\zeta_{2^5})$ which has degree 16 over $\mathbb{Q}$.

3.19 Lemma. Suppose $f \in K[t]$ is irreducible, where K has characteristic 0. For any even degree ≤ 30 , the following is an upper bound on $d(f)$:

$\deg(f)$	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30
$d(f) \leq$	1	2	2	3	3	3	2	4	4	4	2	4	3	3	4

Proof. Let $n = \deg(f)$. Since K has characteristic 0 and f is irreducible, the roots of f are distinct. Considering the action of $\text{Gal}(f)$ on the n roots, we get $\text{Gal}(f) \leq S_n$ and moreover, $\text{Gal}(f)$ acts transitively on $\{1, 2, \dots, n\}$. Using the classification of transitive permutation groups in low degrees [HR, Table 1], we used [GAP] to compute $d(G)$

for each of them and took maximums to get the values in the table. Here $d(G)$ can easily be computed from the abelianization of G , and doing so for all 37,256 relevant transitive groups takes about a minute. $\square$

3.20 Quotients of nonprime order. So far, we have considered $\chi: H \rightarrow \mathbb{Z}/q\mathbb{Z}$ where q is a prime. However, Theorem 3.3 also permits the case of $\chi: H \rightarrow \mathbb{Z}/q^e\mathbb{Z}$ where $e > 1$. Now $\mathbb{Z}/q^e\mathbb{Z}$ is not a field, so to follow the discussion leading to Lemma 3.5, we would need to think about $(\mathbb{Z}/q^e\mathbb{Z})[\mathbb{Z}/m\mathbb{Z}]$ -modules and their attendant quirks. To sidestep this issue for small enough m and q^e , one can work directly with a triangulation of B_m and an associated presentation of $\pi_1 B_m$. One then uses this presentation to describe H and find all $\chi: H \rightarrow \mathbb{Z}/q^e\mathbb{Z}$ directly. One then enumerates all possible metabolizers $A_{(q)}$ of $H_{(q)}$ that satisfy Lemma 3.7 and sees whether each can be ruled out by some χ . This direct approach is much less computationally efficient than in the field case where we can exploit the decomposition of $\bar{H}$ into irreducibles. However, we were still able to use it to obstruct 3833 examples where we had no success using just χ taking values in fields. In those cases, $m = 2$ and $q = 3, 5, 7$ or $(m, q) \in \{(3, 2), (3, 7)\}$. In fact, $(2, 3)$ worked in 3380 cases and $(3, 2)$ worked in 438 cases, with the other values of (m, q) accounting for just 15 cases. (If one knows the $\mathbb{Z}/m\mathbb{Z}$ -action on H , then one could restrict to the possible metabolizers that are $\mathbb{Z}/m\mathbb{Z}$ -invariant. We did not take the care necessary to do so, especially as when $m = 2$ the action is just by ± 1 and hence adds no restrictions.)

3.21 Effective methods and parameters. As stated in Theorem 3.1, in total we showed that 2,211,761 (57.1%) knots in $\mathcal{PS}_{19}$ are not topologically slice using an HKL test. The original Lemma 3.5 accounted for the vast majority, namely 2,135,384 (55.2%). Theorem 3.8 or Proposition 3.15 was used for 72,544 (1.9%) knots, and the method of Section 3.20 was used for 3833 (0.03%) knots.

For the successful obstructions, the degree m of the cover used was primarily 2 (84.1%), 3 (10.6%), 5 (2.6%), 7 (1.4%), 11 (0.8%), and 31 (0.3%), with the remaining values of m accounting for 0.2% of the successful HKL tests; the largest m was 71. The value of q used had median 7, mean 14.3, and $\sigma = 13.6$. The largest q was 661, though 99.6% had $q < 50$. We used $q = 2$, and hence Proposition 3.15, for 67,025 knots, which is 3.0% of the successful HKL tests.

3.22 Implementation. Our implementation of these topological slice obstructions was incorporated into the software package SnapPy [CDGW] and is available when one uses it in SageMath [Sage]. The initial version was released in 2021 as part of SnapPy 3.0 and was restricted to the test of Lemma 3.5, and also required m to be prime (not just a prime power); this is the version referred to in [MP, OS]. An enhanced version that uses Theorem 3.8, allows m to be a prime power, and uses Proposition 3.15 when $q = 2$ will be part of SnapPy 3.3 and is included in [DG]. One

difference compared to [HKL] is that we use the default simplified presentation for $\pi_1 E_K$ that SnapPy provides, rather than a Wirtinger presentation coming from a knot diagram; the former has many fewer generators but much longer relators.

3.23 Possible refinements. We end this section by outlining two possible refinements. Given how effective this technique is overall, we expect either would likely obstruct topological sliceness of some of the ≈ 1400 knots where that property is unknown.

First, one could greatly reduce the number of metabolizers to be ruled out by computing the linking form lk on $H_1(B_m; \mathbb{Z})$. To illustrate the idea, consider the common situation where $m = 2$ and $H_{(q)} = (\mathbb{Z}/q\mathbb{Z})^2 = \bar{H}$. Here, the generator of $\mathbb{Z}/2\mathbb{Z}$ acts on all of $\bar{H}$ by $v \mapsto -v$, so $\bar{H}$ decomposes as V_1^2 where $\dim V_1 = 1$. To apply Theorem 3.8, we will need to strongly obstruct V_1^2 , and so we need some $q + 1$ different Δ_χ to not be norms. Now lk must be the standard hyperbolic bilinear form on $\bar{H}$, that is, there is a basis $\{x, y\}$ of $\bar{H}$ with $lk(x, x) = lk(y, y) = 0$ and $lk(x, y) = 1/q$ (see e.g. Chapter 7 of [Asc], specifically (21.2) and note $\bar{H}$ must have Witt index 1 as $\dim \bar{A} = 1$). Thus, $\langle x \rangle$ and $\langle y \rangle$ are the only possibilities for $\bar{A}$ and so we need only check two Δ_χ . (A caveat is that many ribbon knots in this situation have all Δ_χ being nonnorms, in which case Theorem 3.8 would succeed regardless.) Direct computation of lk from a triangulation was done in [BB] using Regina [BBP⁺]. When m is small, Regina succeeds in computing lk for the B_m that are of interest here. However, it would require delicate book-keeping to “line up” the answer for lk with the computation of Δ_χ , so we do not pursue this here. (Another approach to find lk would be to use [Nos], which requires a Seifert surface.)

Second, one can often determine enough of lk to greatly restrict the possible $\bar{A}$ just from the decomposition of $\bar{H}$ into irreducibles. For an $\mathbb{F}_q[\mathbb{Z}/m\mathbb{Z}]$ -module V , let V^* be its dual or contragredient; in coordinates, if $\psi: \mathbb{Z}/m\mathbb{Z} \rightarrow \mathrm{GL}_n \mathbb{F}_q$ describes V then $\psi^*(g) = (\psi(g^{-1}))^t$. If V and W are irreducible invariant subspaces of $\bar{H}$, we get a map $V \rightarrow W^*$ of $\mathbb{F}_q[\mathbb{Z}/m\mathbb{Z}]$ -modules by $v \mapsto lk(v, \cdot)$; by Schur’s lemma, this implies that V and W are orthogonal unless $V \cong W^*$. Suppose that $\bar{H}$ decomposes as $\bigoplus_{i=1}^k V_i^{e_i}$ where no V_i is self-dual and so isomorphic to V_i^* . Then lk vanishes on each $V_i^{e_i}$, and we can rewrite the decomposition as an *orthogonal* sum $\bigoplus_{i=1}^{k/2} W_i$ where $W_i = V_i^{e_i} \oplus (V_i^*)^{e_i}$ and nondegeneracy forces each nonzero $v \in V_i$ to pair nontrivially with some $w \in V_i^*$. For example, if $\bar{H} = V_1 \oplus V_1^* \oplus V_2 \oplus V_2^* \oplus V_3 \oplus V_3^*$ where all V_i have the same dimension, then Theorem 3.8 requires us to find nonnorm Δ_χ for four of these factors to obstruct slicing. However, we claim that if there are nonnorm Δ_χ for both V_i and V_i^* for a single i then the knot cannot be topologically slice. This is because any $\bar{A}$ will have to meet $V_i \oplus V_i^*$ in $\{0\}$, V_i , or V_i^* , and so $\bar{A}$ is in the kernel of at least one of the two χ .

$\#K \in \mathcal{PS}_{19}$	$\#K \in \mathcal{HKL}^c$	smooth slice obstruction
195,155	6776	One of the s -invariants $\{s_{\mathbb{Q}}, s_{\mathbb{F}_2}, s_{\mathbb{F}_3}\}$
60,929	2998	The Steenrod square $s^{\text{Sq}_o^1}$ on Kh_{odd}^*
890	838	The LEO invariant $\tilde{s}_c$
6	6	An s -invariant for the $\mathfrak{sl}_3$ -version of Kh^*
76,885	13	Goeritz bifactor. test for alternating knots
333,912	10,631	total distinct knots obstructed

Table 9. Above is the breakdown of which smooth slice obstructions were used to prove Theorem 4.1. The first column gives the number of knots in $\mathcal{PS}_{19}$ obstructed by each invariant. For the second column, $\mathcal{HKL}^c \subset \mathcal{PS}_{19}$ is the complement of the knots shown not to be topologically slice by Theorem 3.1; hence this column represents the new information gleaned from Theorem 4.1. Each knot is counted only once, for the invariant that obstructs it that is highest on the table. The LEO invariant $\tilde{s}_c$ necessarily obstructs anything handled by $\{s_{\mathbb{Q}}, s_{\mathbb{F}_2}, s_{\mathbb{F}_3}, s^{\text{Sq}_o^1}\}$, for a total of 256,974 knots in $\mathcal{PS}_{19}$ and 10,612 knots in $\mathcal{HKL}^c$.

4 Smooth slice obstructions

We computed many different smooth slice obstructions coming from Khovanov homology, knot Floer homology, and gauge theory, yielding:

4.1 Theorem. *Of the knots in $\mathcal{PS}_{19}$, some 333,912 (8.6%) are obstructed from being smoothly slice by the invariants listed in Table 9.*

4.2 Remark. Note that Theorem 3.1, which uses the topological HKL tests, covers 6.6 times more knots than Theorem 4.1. Moreover, Theorem 4.1 obstructs only 10,631 knots (0.3% of $\mathcal{PS}_{19}$) that are not included in Theorem 3.1.

The set of invariants used to prove Theorem 4.1 were:

- (1) Rasmussen's s -invariant [Ras], over the fields $\mathbb{Q}$, $\mathbb{F}_2$, and $\mathbb{F}_3$. We denote these by $s_{\mathbb{Q}}$, $s_{\mathbb{F}_2}$, and, $s_{\mathbb{F}_3}$, each of which takes values in $\mathbb{Z}$ and is 0 on all smoothly slice knots.
- (2) The Sq_o^1 -invariant in odd Khovanov homology from [SSS, §5.6]. We denote it by $s^{\text{Sq}_o^1}$, and it takes values in $\mathbb{Z}^4$ and is 0 on all smoothly slice knots.
- (3) The invariant $\tilde{s}_c$ from [DLS, §7], which is defined in terms of local equivalence classes of local even-odd (LEO) triples that mix together the original and odd

flavors of Khovanov homology. This takes values in $\mathbb{Z}^2$ and is 0 on all smoothly slice knots.

- (4) Various s -invariants associated to the $\mathfrak{sl}_3$ -Khovanov homology [Sch5]. These are quite difficult to compute, so we did so only for 15,355 knots, including all those unresolved by other methods.
- (5) For alternating knots, we employed the criterion of [LM, Lemma 3], which involves bifactorability of the Goeritz matrices and is based on Donaldson's theorem.

All computations of the Khovanov-based invariants in (1–4) were done using Schütz's KnotJob [Sch2]. By Section 6 of [DLS], the LEO invariant $\tilde{s}_c$ is a stronger smooth slice invariant than any of $s_{\mathbb{Q}}, s_{\mathbb{F}_2}, s_{\mathbb{F}_3}, s^{\text{Sq}^1}_0$; that is, if any of those four invariants obstruct smooth slicing, so does $\tilde{s}_c$. For alternating knots in $\mathcal{PS}_{19}$, the obstruction (5) specific to that setting was the only one of (1–5) that obstructed smooth slicing. By [DLS, Prop. 6.30], none of (1–3) can obstruct smooth slicing for an alternating knot with signature 0, so this is to be expected.

Other invariants we looked at included:

- The invariants τ , ν , and ϵ from knot Floer homology, computed via [Sza]. These obstructed essentially the same set of knots as did $(s_{\mathbb{Q}}, s_{\mathbb{F}_2}, s_{\mathbb{F}_3})$. Specifically, every knot obstructed by one of (τ, ν, ϵ) was also obstructed by one of $(s_{\mathbb{Q}}, s_{\mathbb{F}_2}, s_{\mathbb{F}_3})$, with the converse holding 99.96% of the time.
- The Sq^1 -invariant from the original Khovanov homology [LS] obstructs exactly the same knots in $\mathcal{PS}_{19}$ as $(s_{\mathbb{Q}}, s_{\mathbb{F}_2}, s_{\mathbb{F}_3})$. The same is true for the $s^{\mathbb{Z}}$ -invariant of [Sch4]. In general, neither of these invariants can give more information than the LEO invariant $\tilde{s}_c$ [DLS].
- The higher Sq^2 -invariant of [LS] is very difficult to compute for a 19 crossing knot, and the analogues for odd Khovanov homology from [Sch6] are even harder. (This is because there is no known Bar-Natan-style “scanning algorithm” for these invariants; compare [Sch1], which applies even to $\tilde{s}_c$.) While these invariants can provide slice obstructions not seen by $\tilde{s}_c$, we only computed Sq^2 from [LS] for about 2700 knots where all the other Khovanov invariants vanished. Unfortunately, Sq^2 also vanished for these knots. The same was true for the roughly 130 knots where we computed the versions of Sq^2 for Kh^*_{odd} .

5 Zero-friends

The 0-surgery on a knot K in S^3 is the unique Dehn surgery where $b_1 > 0$. A pair of knots in S^3 are *0-friends* when their 0-surgeries are homeomorphic. In this section, we study some 0-friends of the knots in $\mathcal{PS}_{19}$. The motivation for this is twofold. First, and most tantalisingly, if K and K' are 0-friends where K is smoothly slice and K' is not, then there is an exotic smooth structure on S^4 , disproving the smooth 4-dimensional Poincaré Conjecture; see [MP, §1]. Second, even with that conjecture unresolved, in favorable circumstances one can show that certain 0-friends must have the same smooth slice status; for example, this was used to show that the Conway knot is not smoothly slice [Pic]. In Section 5.6, we use 0-friends to determine the smooth slice status of 25 knots in $\mathcal{PS}_{19}$. Section 5.13 gives 4 pairs of 0-friends where we can only determine smooth sliceness of one of the knots; these are thus candidates for constructing exotic smooth 4-spheres.

5.1 Finding 0-friends. There is an algorithm to determine when two 3-manifolds are homeomorphic (see e.g. [Kup]). While the known algorithms are unimplementable, in practice one can still usually decide this, especially in the presence of hyperbolic geometry [HW]. With 114 exceptions (0.003%), the knots in $\mathcal{PS}_{19}$ have hyperbolic 0-surgeries, so the first question is how many pairs of 0-friends there are in $\mathcal{PS}_{19}$ itself. The answer is just 101 pairs.¹ This is consistent with [AKW], which found only 6 such pairs among all $\approx 313,000$ prime knots with ≤ 15 crossings. Note also that the 0-friend used in [Pic] is much more complicated than the Conway knot itself. Therefore, to usefully study 0-friends of knots in $\mathcal{PS}_{19}$, we need a technique that can generate them from a single given knot K .

We used the following method from [DOR, §9.3] to generate more than 500,000 0-friends of the knots in $\mathcal{PS}_{19}$; we explain in Section 5.4 why this should generate most of the interesting small-to-medium 0-friends. One example pair of 0-friends is shown in Figure 4.

- (1) For each knot K whose 0-surgery $Z_K = S^3_0(K)$ is hyperbolic, we looked at all closed geodesics γ of length at most 3 that generated $H_1(Z_K; \mathbb{Z}) \cong \mathbb{Z}$, using [HW, CDGW]. (The reason we stopped at length 3 is explained in Section 5.4.)
- (2) For each γ , we considered the exterior $X = Z_K \setminus \gamma$, which must itself be hyperbolic. The hyperbolic structure on X determines a finite list of Dehn fillings on

¹The common 0-surgery is hyperbolic for 98 of these 101 pairs of 0-friends. There are only two nontrivial groups of mutual 0-friends, both with four knots: $\{K11n97, 16n86947, 18nh_{00000399}, 18nh_{00000400}\}$ and $\{K11n67, 18nh_{00000254}, 19nh_{000000631}, 19nh_{000000632}\}$. See [DG] for all 186 knots involved.

it that might be S^3 , and this can be used to identify whether X is the exterior of some unknown knot K' in S^3 . (See e.g. [Dun] for more on this.)

- (3) For each knot exterior X identified in Step 2, we used [DOR] to find an actual knot diagram for K' .

5.2 Remark. This method uses numerical computations about the hyperbolic structures that are not rigorous. However, each pair of 0-friends in our data must be genuine since filling, drilling, and finding knot diagrams are all combinatorial operations that are implemented at the level of manipulating triangulations of the relevant 3-manifolds. Such numerical issues instead mean that our list of 0-friends may not be complete according to the above description; still, we expect it contains 99% of such 0-friends.

5.3 Overview of 0-friends. We identified at least one 0-friend for 546,717 (14.1%) of the knots in $\mathcal{PS}_{19}$. (For these knots, we found an average of 1.5 0-friends, though we were not careful about avoiding duplicates.) The crossing number of the diagram of the 0-friend K' had mean 225.7 and median 154, with standard deviation σ of 219.2. (We make no claim that these are minimal crossing diagrams, though we suspect very few can be simplified to reduce the number of crossings by even 50%.) This makes most of these diagrams much too large to allow computing Khovanov homology and hence most of the invariants of Section 4. In contrast, the exteriors $E_{K'}$ are only modestly more complicated than the original E_K . The number of tetrahedra in the triangulation of $E_{K'}$ had mean and median 39.0 as compared to 31.0 for E_K , an increase of only 25% (here $\sigma = 7.91$ and $\sigma = 3.84$ respectively).

In the end, we were able to determine the smooth slice status for 78,507 0-friends of 56,319 knots in $\mathcal{PS}_{19}$; we denote these 0-friend knots by $\mathcal{ZF}$ and show information about their crossing number and hyperbolic volume in Figure 12. Some 82.0% of the knots in $\mathcal{ZF}$ are ribbon and 18.0% are not smoothly slice. (For all the knots in $\mathcal{PS}_{19}$ where we found a 0-friend, we had 71.5% ribbon and 28% not smoothly slice, with 0.4% unknown; this contrasts with $\mathcal{PS}_{19}$ as a whole where 42.2% are ribbon, 57.5% are not smoothly slice, and 0.3% unknown.) Initially, based on the computations described in Sections 2–4, there were 36 knots in $\mathcal{PS}_{19}$ of unknown smooth slice status where this was determined for a 0-friend. For 28 of these knots in $\mathcal{PS}_{19}$, we can use the 0-friend to determine the smooth slice status of the original; see Theorem 5.10 for more and Section 5.13 for the 8 mysterious remaining knots.

5.4 Hyperbolic biases. Our approach in Section 5.1 may seem to be one of convenience: we restrict to when Z_K is hyperbolic and then using that hyperbolic structure to guide the search for the 0-friend.

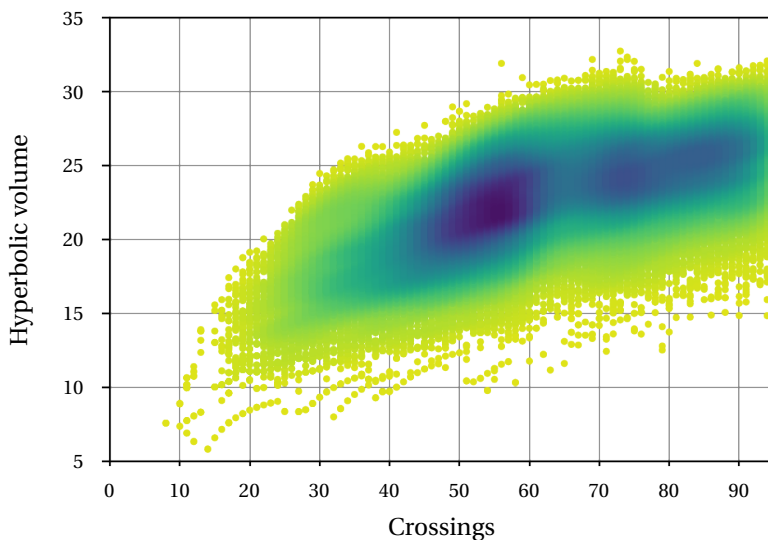


Figure 12. For each 0-friend K' in $\mathbb{ZF}$, this plot compares the crossing number of the simplest known diagram for K' with the volume of the hyperbolic structure on $E_{K'}$. The latter has mean 22.4, median 22.8, and $\sigma = 3.5$. This is about 9% more than the corresponding original knots in $\mathcal{PS}_{19}$; compare also Figure 2.

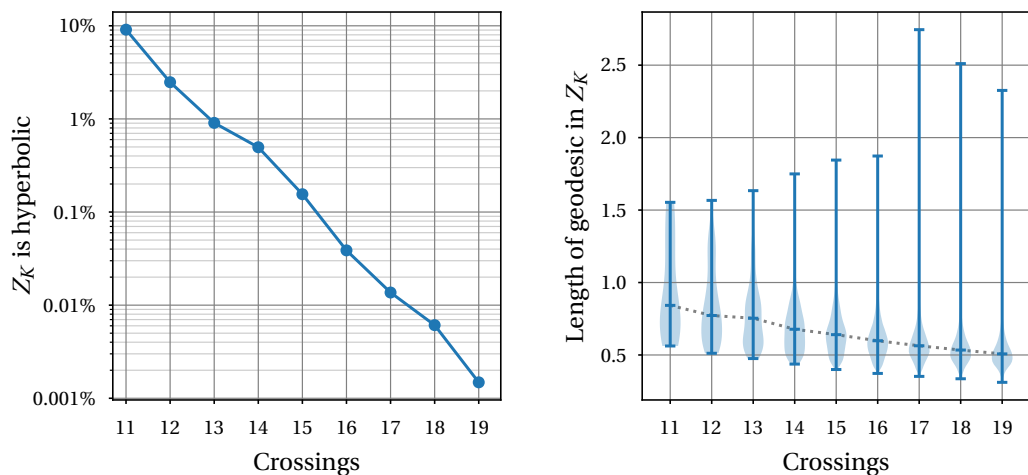


Figure 13. There are 93 knots K in $\mathcal{PS}_{19}$ whose exterior E_K is hyperbolic but Z_K is not hyperbolic. At left, we see the portion of such exceptional knots declines exponentially in the number of crossings, ending at 0.0015% (1 in 67,500) for 19-crossing knots. For $\mathcal{PS}_{19}$, every time Z_K was hyperbolic the core of the Dehn surgery solid torus was isotopic to a geodesic. The violin plot at right shows how the length of that geodesic changes as the number of crossings increases; the dotted line passes through the medians of each sample and the top and bottom lines are the maxes and mins.

We already noted that Z_K is hyperbolic for all but 114 knots K in $\mathcal{PS}_{19}$; this includes the 21 knots in $\mathcal{PS}_{19}$ where E_K itself is not hyperbolic. Moreover, the portion of such exceptional knots appears to decline exponentially in the number of crossings, as shown in the left plot of Figure 13. Thus insisting that Z_K be hyperbolic requires that we skip very few knots.

However, while any K' that is a 0-friend of K gives a corresponding knot in Z_K that generates $H_1(Z_K; \mathbb{Z})$, in Step 1 we restrict to those knots in Z_K that are *isotopic to geodesics* in its hyperbolic structure. Remarkably, whenever Z_K is hyperbolic for K in $\mathcal{PS}_{19}$, the core of the Dehn surgery solid torus in Z_K is indeed isotopic to a geodesic. We know of no geometric heuristic that suggests this pattern will not continue; if anything, we would expect it to become more marked if that were possible. Thus, when looking for a 0-friend, considering just the exteriors of geodesics in Z_K seems to be a very mild restriction, or possibly not one at all!

It remains to explain why we chose to limit our search to geodesics in Z_K of length at most 3. For the whole of $\mathcal{PS}_{19}$, the length of the core geodesic in Z_K had mean 0.54

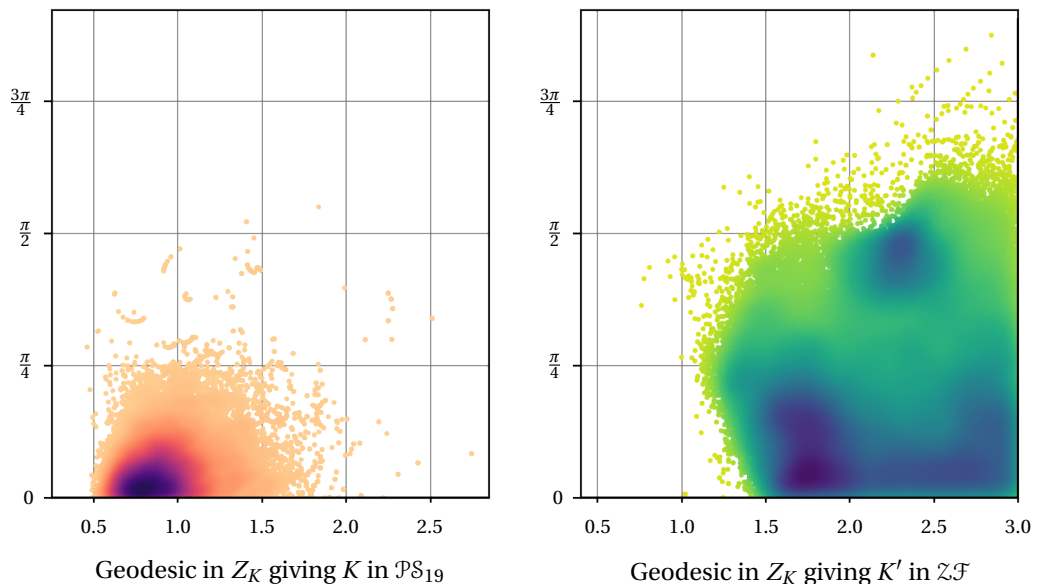


Figure 14. A closed geodesic γ in a hyperbolic manifold M has a length $L \in \mathbb{R}_{>0}$ and also a twist $\theta \in \mathbb{R}/(2\pi\mathbb{Z})$ recording the holonomy of going around γ ; these are combined into its *complex length* $L + i\theta \in \mathbb{C}/(2\pi i\mathbb{Z})$. The above shows the complex lengths for the corresponding geodesics in Z_K for each pair of 0-friends K in $\mathcal{PS}_{19}$ and K' in $\mathcal{ZF}$. The complex length depends on the orientation of M but not of γ ; as the original choice of K or $\bar{K}$ in [Bur] is somewhat arbitrary, we have normalized θ modulo π rather than 2π in both cases.

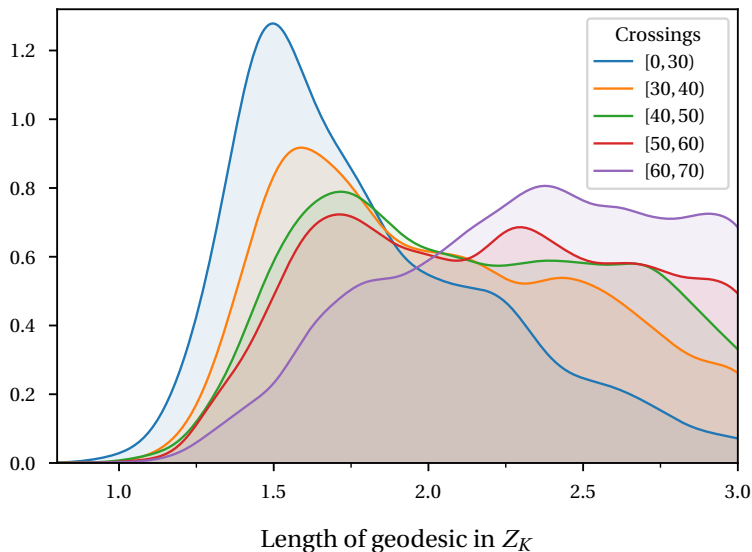


Figure 15. This plot explores how the length of the geodesic in Z_K that gives the 0-friend K' relates to the number of crossings of K' . Specifically, the 0-friends were grouped by crossing as indicated in the upper-right and then a smoothed histogram is shown for the lengths of the corresponding geodesics. The groups had between 2,659 and 25,316 knots each, where the size increases with the crossing number. Here, we include all 0-friends found for knots in $\mathcal{PS}_{19}$, not just those in $\mathcal{ZF}$.

and median 0.51, with $\sigma = 0.11$. No geodesic was longer than 2.75, and fewer than 0.01% were longer than 1.5. Moreover, as the number of crossings increased, the median length decreased as shown in the right plot of Figure 13, though a number of high outliers remain. Thus a cutoff of 3 is at least big enough to capture all of the 0-friends within $\mathcal{PS}_{19}$. On the other hand, the right plot in Figure 14 makes it clear there should be many 0-friends of knots in $\mathcal{PS}_{19}$ coming from geodesics in Z_K of length more than 3. The question is then how many “small-to-medium” 0-friends we are missing, where concretely we mean those with a diagram of at most 50 crossings. Figure 15 suggests that we have found roughly 90% of those with less than 30 crossings, say 75% of those with less than 40 crossings, and perhaps 60% of those with less than 50 crossings. On the other hand, we likely have much less than 50% of those with crossing number in $[50, 70)$.

It would be computationally feasible to raise the cutoff. As the number of geodesics grows exponentially in the length, this gets increasingly difficult, but considering those of length up to 4 (or perhaps 5) should be possible in most of these examples.

5.5 Hyperbolic aside. We next discuss an intriguing pattern in the hyperbolic geometry of these examples; it is not directly related to the main goal of using 0-friends to understand the knots in $\mathcal{PS}_{19}$, so you may want to skip ahead to Section 5.6. Our motivation here was to try to predict in advance which geodesics in Z_K give K' with a diagram with relatively few crossings, but we gained little purchase on this goal.

We start by considering the difference in volume between E_K and Z_K . By Thurston, one always has $\text{Vol}(Z_K) < \text{Vol}(E_K)$, and presuming we are in the situation that E_K is the complement of a closed geodesic γ in Z_K , the difference $\Delta \text{Vol} = \text{Vol}(E_K) - \text{Vol}(Z_K)$ is typically controlled by the length L of γ , with $\Delta \text{Vol} \approx \frac{\pi}{2}L$. The latter is always true asymptotically for sufficiently large Dehn fillings on E_K by [NZ]. Even for smaller Dehn fillings, as long as one can deform the hyperbolic structure on E_K to that of Z_K via hyperbolic cone manifolds, it is usually a good approximation because of Schäfli's formula; see e.g. [AST, §A.3]. Here, this pattern comes through very strongly in Figure 16. While for any knot K with a diagram with c crossings one has $\text{Vol}(E_K) < \nu_{\text{oct}} \cdot c < 3.67c$, there is no converse to this and Figure 12 shows only a weak correlation between $\text{Vol}(E_{K'})$, and the crossing number of the diagram for K' .

Less expected is the relationship between $\text{Vol}(E_K)$ and L shown in Figure 17, namely that $\text{Vol}(E_K)$ is roughly linear in $1/L$. Intriguingly, this holds only for the original K in $\mathcal{PS}_{19}$, not the 0-friends K' in $\mathcal{ZF}$. We now re-frame this phenomenon to remove reference to Z_K . Consider a horotorus T that cuts off a cusp neighborhood C of E_K ; the Euclidean structure on T is unique up to dilation. Let μ and λ be the Euclidean translations corresponding to the meridian and longitude curves on T ; one defines the *cuspid shape* as $\tau_K = \lambda/\mu$, where by convention orientations are chosen so $\text{Im}(\tau_K) > 0$. A simple calculation using Lemma 4.2 of [NZ] suggests that $L \approx \text{Re}(2\pi i/\tau_K)$. (To see this, apply the lemma in the case $p = 0$ and $q = 1$, using that $v(u) = \tau_K u + O(u^3)$ by their equation (12), and $v(u) = 2\pi i$ by their equation (32).) This would imply $2\pi/L \approx |\tau_K|^2/\text{Im}(\tau_K)$, suggesting that we compare $\text{Vol}(E_K)$ with $|\tau_K|^2/\text{Im}(\tau_K)$. Doing so results in the striking plot in Figure 18. This brings to mind [DJLTs], which relates $\text{Re}(\tau_K)$ to σ_K , since plausibly slice knots have $\sigma_K = 0$, but we do not know a direct connection to Figure 18.

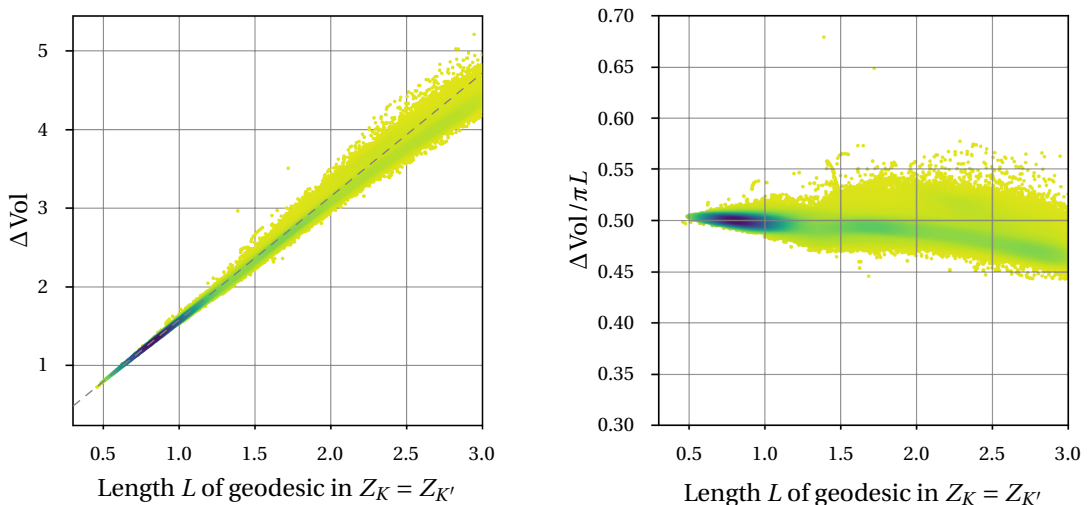


Figure 16. At left, for each K' in $\mathcal{ZF}$, we plot $\Delta \text{Vol} = \text{Vol}(E_{K'}) - \text{Vol}(Z_{K'})$ against the length L of the corresponding geodesic in $Z_{K'}$ and we do the same for its corresponding 0-friend K in $\mathcal{PS}_{19}$. (From Figure 14, we see that typically $L > 1.5$ for K' and $L < 1.5$ for K . The dotted line is the prediction $\Delta \text{Vol} = \frac{\pi}{2}L$. At right, the ratio $\Delta \text{Vol}/\pi L$ is plotted instead; compare this with the figure in [AST, §A.3] which is for much smaller manifolds than those considered here.

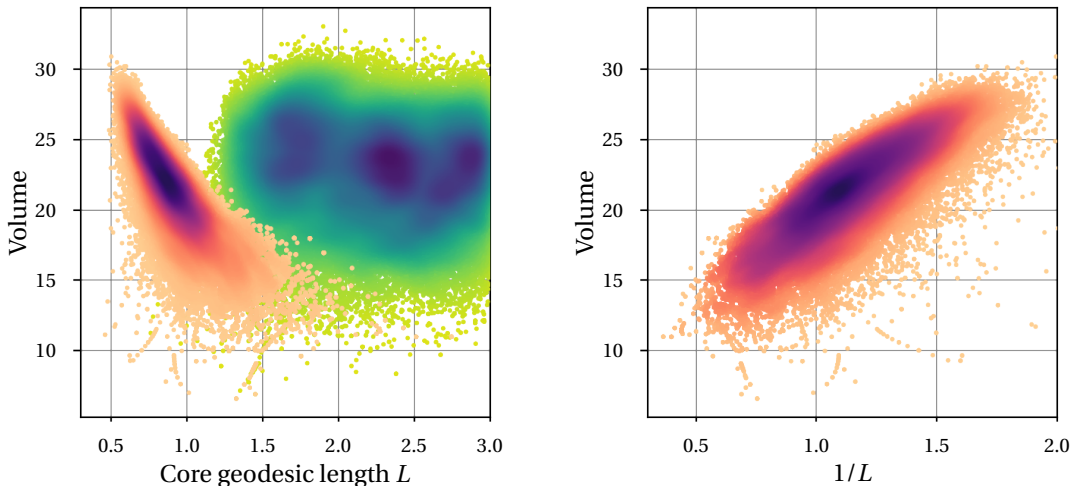


Figure 17. At left, for each pair of 0-friends K in $\mathcal{PS}_{19}$ and K' in $\mathcal{ZF}$, we plot the volume of E_K and $E'_{K'}$ against the length L of the corresponding geodesics in $Z_K = Z'_{K'}$; the orange-red-purple blob where mostly $L \in [0.5, 1.5]$ corresponds to K in $\mathcal{PS}_{19}$ and the green-blue blob where mostly $L \in [1.5, 3]$ corresponds to K' in $\mathcal{ZF}$. Only the $\mathcal{PS}_{19}$ blob shows much structure, and that structure becomes even more evident at right where the x -axis is now $1/L$ rather than L .

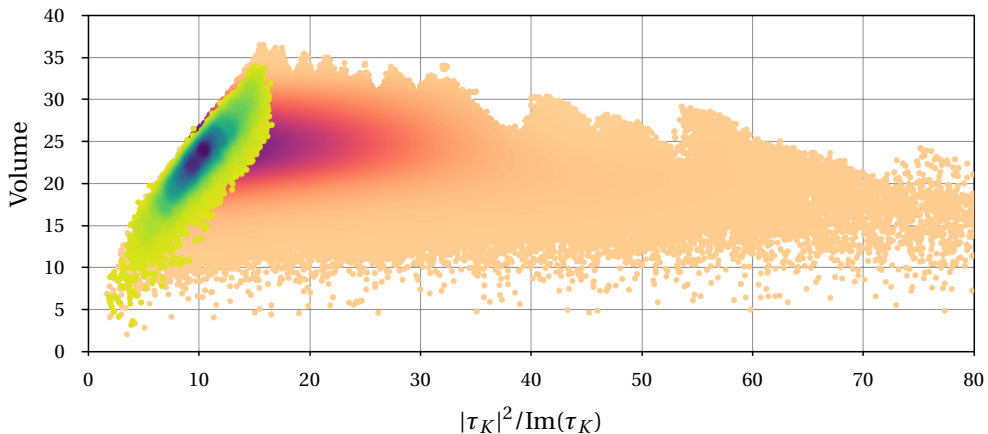


Figure 18. This plot compares the hyperbolic volume of E_K with the shown quantity based on the cusp shape τ_K . The smaller green-blue blob on the left corresponds to hyperbolic knots in $\mathcal{PS}_{19}$ with at most 16 crossings. The larger peach-purple blob corresponds to all hyperbolic knots with at most 16 crossings. The densest part of the larger blob is hidden behind the smaller one.

5.6 Applications. To explain what the 36 exceptional 0-friends say about the corresponding knots in $\mathcal{PS}_{19}$, we need to introduce the framework of red-blue-green (RBG) links, following [MP, §3].

5.7 Definition. An *RBG link* $L = R \cup B \cup G \subset S^3$ is a 3-component link, with rational framings r, b, g respectively, such that there exist homeomorphisms $\psi_B: S^3_{r,g}(R \cup G) \rightarrow S^3$ and $\psi_G: S^3_{r,b}(R \cup B) \rightarrow S^3$ and such that $H_1(S^3_{r,b,g}(L); \mathbb{Z}) \cong \mathbb{Z}$.

An RBG link describes a pair of 0-friends K_B and K_G where K_B is the image of B under ψ_B and K_G is the image of G under ψ_G ; their common 0-surgery is $S^3_{r,b,g}(L)$. Conversely, any pair of 0-friends can be encoded by an RBG link [MP, Theorem 1.2]. An RBG link for the 0-friends in Figure 4 is shown in Figure 19.

We say that an RBG link is *super-special* when the two sublinks $R \cup B$ and $R \cup G$ are both Hopf links and the framings satisfy $b = g = 0$ and $r \in \mathbb{Z}$; this is a subset of the special RBG links of [MP]. The key facts for us are:

5.8 Theorem [MP]. Suppose L is a super-special RBG link. If r is even, then K_B and K_G have homeomorphic traces; if $r = 0$, their traces are diffeomorphic. If instead r is odd and their common 0-surgery has trivial mapping class group, then their traces are not homeomorphic.

Proof. This is in Sections 3 and 4 of [MP]; specifically, combine Theorems 3.7 and 3.13 with Lemmas 4.2 and 4.3, noting Remark 3.8. $\square$

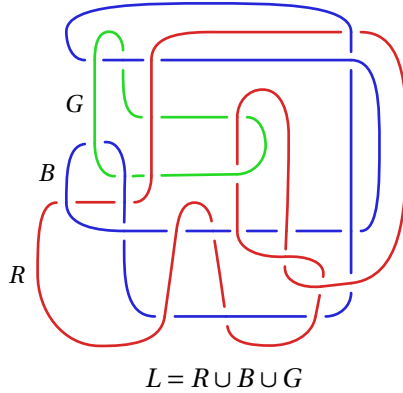


Figure 19. An RGB link L for the 0-friends K and K' in Figure 4. With $r = 1$, $b = g = 0$, one has $K' = K_B$ and $K = K_G$. The link L was found from (K, K') using the method of Section 5.11.

5.9 Theorem [Nak, Ren]. Suppose L is a super-special RGB link. Suppose $s_{\mathbb{F}}(K_G) \neq 0$, where $s_{\mathbb{F}}$ is the Rasmussen s -invariant over a field $\mathbb{F}$. Then both K_B and K_G are not smoothly slice.

Proof. As $s_{\mathbb{F}}(K_G) \neq 0$, of course K_G is not smoothly slice. For K_B , we use Theorem 3.13 of [Nak]; this result is conditional on Conjecture 2.15 in that paper, but this was established as Corollary 1.5 of [Ren]. If K_B was smoothly slice, since R is the unknot, Theorem 3.13 of [Nak] applies to say $s_{\mathbb{F}}(K_G) = 0$, a contradiction. $\square$

The main result of this section is:

5.10 Theorem. The 25 knots listed in Table 10 are not smoothly slice.

The first two knots in Table 10 were already known not to be smoothly slice by [Pic] in the case of the Conway knot $K11n34$ and by unpublished work of Ciprian Manolescu, Lisa Piccirillo, and Julius Zhang in the case of $K13n866$ using a similar method to [Pic].

Proof of Theorem 5.10. For each of the 25 knots in the statement, the method of Section 5.11 below was used to find a super-special RGB link that encoded the relationship between the original K in $\mathcal{PS}_{19}$ and its 0-friend K' ; this is recorded in Table 10.

For $K = 18nh_{00010270}$, as $r = 0$ in that RGB link we learn K and K' have diffeomorphic traces by Theorem 5.8. By the Trace Embedding Lemma (see e.g. [MP, Lemma 3.5]), it follows that K and K' have the same smooth slice status. Now K' is not smoothly slice, as $s^{\text{Sq}_o(K)}(K') \neq 0$, and so K cannot be either. The remaining 24 knots in Table 10 are covered by Theorem 5.9. $\square$

knot	DT code of RBG link	r	K' notes
$K11n34$	vc demhUSFVDCoKIMeJaRgtNpBql	0	$S_{\mathbb{F}_3} = 2$
$K13n866$	tc biisTAb rNEomgQKhFiLdJpc	-1	$S_{\mathbb{F}_3} = 2$
$K15n25044$	xcdhlUkRJgNx sVpQMLFWhdBCoAlTe	-4	$S_{\mathbb{F}_3} = 2$
$16n180537$	rc bdlGCARMoFBkQhiEPIDNJ	1	$S_{\mathbb{F}_3} = 2$
$16n74539$	tc biiEnTOBDlRpqMfKgaCs jHi	-1	$S_{\mathbb{F}_3} = -2$
$17nh_{0001844}$	sc bfkjFinLRAQObhDpeSKmGc	1	$S_{\mathbb{F}_3} = 2$
$17nh_{0002715}$	xcd oelHtPXVRDuSEaqFwCGmOibKnJ	0	$S_{\mathbb{F}_3} = -2$
$17nh_{0212094}$	sccdlrcesbjkOGLmifpqHnda	-1	$S_{\mathbb{F}_3} = 2$
$17nh_{0212095}$	qccd jenaQcmPOGldfjbkHI	-1	$S_{\mathbb{F}_3} = 2$
$18nh_{00010270}$	zcdkkgNKtZYuaQpCs rVDjXmwbOhlIFE	0	$S_{\mathbb{F}_3}^{\text{Sq}^1(K)} \neq 0$
$18nh_{00166702}$	rc ehemldoJR hqpEIFnabcgK	0	$S_{\mathbb{F}_3} = -2$
$18nh_{00610378}$	tc celDPAECTJSQGMNrKfHBIIO	1	$S_{\mathbb{F}_3} = 2$
$18nh_{00610381}$	uc ec mKDGBAIRCOSehTpqInFuMj	-1	$S_{\mathbb{F}_3} = 2$
$19nh_{000002588}$	yc blk uLFkRYOmWeSpAvNThIxCbGqDJ	-1	$S_{\mathbb{F}_3} = -2$
$19nh_{000003154}$	rc gfeClHoJp dBNqFaRMGkEi	3	$S_{\mathbb{F}_3} = -2$
$19nh_{000003570}$	uc iffKIptSBhoQUNrEdgCflamJ	-5	$S_{\mathbb{F}_3} = 2$
$19nh_{000032808}$	xchigIfEOtUkwXmAsPjCVhQGdbNLR	0	$S_{\mathbb{F}_3} = -2$
$19nh_{000076489}$	wcbjkDftqRSMWovpGLKujcbEAhiN	-1	$S_{\mathbb{F}_3} = -2$
$19nh_{000018991}$	wcdijMJrFTLWi qgNUDaSvBhPkEoC	0	$S_{\mathbb{F}_3} = -2$
$19nh_{000066839}$	xc bnhMqXegIdTFOPcBrSWlavUHnJK	-1	$S_{\mathbb{F}_3} = -2$
$19nh_{000066841}$	yc bohEgYwUtsRxpMoJLVkiCafbNdqh	1	$S_{\mathbb{F}_3} = -2$
$19nh_{000177115}$	zcdnhRqPoGwUVasCnTxyJzibDMEHflk	0	$S_{\mathbb{F}_3} = 2$
$19nh_{000177116}$	zcdscVuopGhWmqLZrFedcsiknXyTJaB	0	$S_{\mathbb{F}_3} = -2$
$19nh_{001336127}$	yc ejjDCNAYSLrXtqWvPBOhkMuifGJE	-1	$S_{\mathbb{F}_3} = -2$
$19nh_{002457201}$	yc dkjOPHexuMLQyrcGVWASKIBNfTDj	0	$S_{\mathbb{F}_3} = -2$

Table 10. Some 0-friends that admit super-special RBG links. In each case, we can conclude that the original knot in $\mathcal{PS}_{19}$ is not smoothly slice.

5.11 Finding RBG links. Given 0-friends K and K' whose exteriors E_K and $E_{K'}$ are hyperbolic, we used the following method to search for a super-special RBG link L that expressed this fact. As in Section 5.1, we will use [DOR] to recover a diagram for L from its exterior as the very last step. In particular, we searched for a 3-cusped hyperbolic manifold Y with cusps labelled red, blue, and green with peripheral framings (μ_R, ν_R) , (μ_B, ν_B) , and (μ_G, ν_G) respectively so that:

- (a) The Dehn filling $Y(\mu_R, \mu_B, \mu_G)$ is homeomorphic to S^3 .
- (b) We have $Y(\nu_R, \cdot, \nu_G) \cong E_K$ by a homeomorphism that takes (μ_B, ν_B) to the standard framing (μ_K, λ_K) for the knot exterior E_K . Likewise, $Y(\nu_R, \nu_B, \cdot) \cong E_{K'}$ where $(\mu_G, \nu_G) \mapsto (\mu_{K'}, \lambda_{K'})$. (Note this implies $Y(\nu_R, \nu_B, \nu_G)$ is the common 0-surgery of K and K' .)
- (c) Both the Dehn fillings $Y(\cdot, \mu_B, \cdot)$, and $Y(\cdot, \cdot, \mu_G)$ have fundamental group $\mathbb{Z}^2$. (This gives the Hopf link condition.)
- (d) The curve ν_B is 0 in $H_1(Y(\mu_R, \cdot, \mu_G); \mathbb{Z})$ and ν_G is 0 in $H_1(Y(\mu_R, \mu_B, \cdot); \mathbb{Z})$. (This gives that $b = g = 0$; note that $r \in \mathbb{Z}$ is automatic since (μ_R, ν_R) is a framing for the red cusp.)

You can check that these four conditions are equivalent to the cores of the fillings in $Y(\mu_R, \mu_B, \mu_G)$ being the needed super-special RBG link. Our method for searching for such Y was:

- (1) We looked at simple curves in the dual 1-skeleton of the triangulation of E_K that were not null-homotopic. For each such curve, we drilled it out to get a 2-cusped manifold X . The cusp of X coming from E_K will be called blue and the new one green. The blue cusp inherits a framing (μ_B, ν_B) from the framing (μ_K, λ_K) for E_K .
- (2) Check if the filling $X(\nu_B, \cdot)$ is homeomorphic to $E_{K'}$. If so, give the green cusp of X the framing (μ_G, ν_G) inherited from the framing $(\mu_{K'}, \lambda_{K'})$ of $E_{K'}$ and proceed to the next step. Otherwise, go back to Step 1. If we succeed, X will be $Y(\mu_R, \cdot, \cdot)$.
- (3) For X passing Step 2, we look through simple curves in the dual 1-skeleton to its triangulation. For each, we drilled it out to get a 3-cusped manifold Y . On the new “red” cusp, we let ν_R (sic) be the curve so that filling along it gives X . Provided that requirement (c) holds, we consider $Y(\cdot, \mu_B, \mu_G)$ and test the finitely-many hyperbolically plausible slopes μ_R to see if any $Y(\mu_R, \mu_B, \mu_G)$ is S^3 . If so, we check whether (d) is satisfied.

- (4) If we succeeded in Step 3, apply [DOR] to get a link L in S^3 whose complement in Y where (μ_R, μ_B, μ_G) are the usual meridians for L . This is the needed super-special RBG link.

Note that this method is not symmetric in K and K' , and sometimes it was necessary to switch their roles to find L .

5.12 Remark. As described in Section 5.1, our 0-friend pairs each come from a pair of simple closed geodesics in the hyperbolic structure of their common 0-filling Z_K . At least assuming these geodesics are disjoint, one could search for the red knot among other closed geodesics in Z_K . Our experience was that this approach is not very effective, and it was often necessary to consider the above larger pool of candidates for the red knot. Moreover, unlike Section 5.4, we know of no heuristic reason why the method of Section 5.11 should typically suffice to find such an L , even though it did in all 36 of the examples at hand.

5.13 Remaining mystery 0-friends. We now turn to an expanded version of Theorem 1.12 from the intro.

5.14 Theorem. *If $18nh_{00000601}$ is not smoothly slice, or if any of $\{16n68278, 17nh_{0010647}, 18nh_{00098198}\}$ are smoothly slice, then there is an exotic smooth 4-sphere. All four knots are topologically slice. The knot $18nh_{00000601}$ is moreover smoothly slice in a homotopy 4-ball.*

Proof. If the hypothesis holds, then from Table 11 we have a pair of 0-friends where one is smoothly slice and the other is not. This gives an exotic smooth 4-sphere as discussed in e.g. [MP, §1]. To sketch the idea, suppose K and K' are 0-friends where K bounds a smooth 2-disk D in D^4 and K' is not smoothly slice. Let $A = D^4 \setminus \nu(D)$ and B be $S^3 \times I$ with a 0-framed 4-dimensional 2-handle attached to $K' \times \{1\}$. Then ∂A and one component of ∂B are both $Z_K \cong Z_{K'}$; gluing these together gives a manifold W with $\partial W = S^3$. Now W is a homotopy 4-ball and $K' \subset \partial W$ bounds a disk in W , basically the core of the 2-handle used to build B . As K' is not smoothly slice, W is an exotic smooth 4-ball, as needed.

The three knots $\{16n68278, 17nh_{0010647}, 18nh_{00098198}\}$ are topologically slice since they all have $\Delta_K = 1$ [FQ, §11.7]. For $K' = 18nh_{00000601}$, we know it is a 0-friend of a smoothly slice knot K . By the preceding paragraph, K' is smoothly slice in a homotopy 4-ball W . Since every homotopy 4-ball is *topologically* standard [FQ], it follows that K' topologically slice in D^4 , as needed. $\square$

knot	DT code of RBG link	r	K' notes
$18nh_{00000601}$	ycjkdnhQyUtMsaVweFIRCXOBgLJDkP	1	ribbon
$16n68278$	tcbjhTDJBgQmsOLhPcekIRFnA	1	$\tilde{s}_c \neq 0$
$17nh_{0010647}$	scdifSmJkOBcNQPFrdAHEIgl	-1	$\tilde{s}_c \neq 0$
$17nh_{0010647}$	xcdhlXqLTCvumPWsoAikRHbnEDfjg	-1	$\tilde{s}_c \neq 0$
$18nh_{00098198}$	rcbgiHMRfPlnJAIOdBcqEKg	-1	$\tilde{s}_c \neq 0$

Table 11. Five more 0-friends with super-special RBG links, where the original knot in $\mathcal{PS}_{19}$ remains a mystery. The two K' for $17nh_{0010647}$ are distinct.

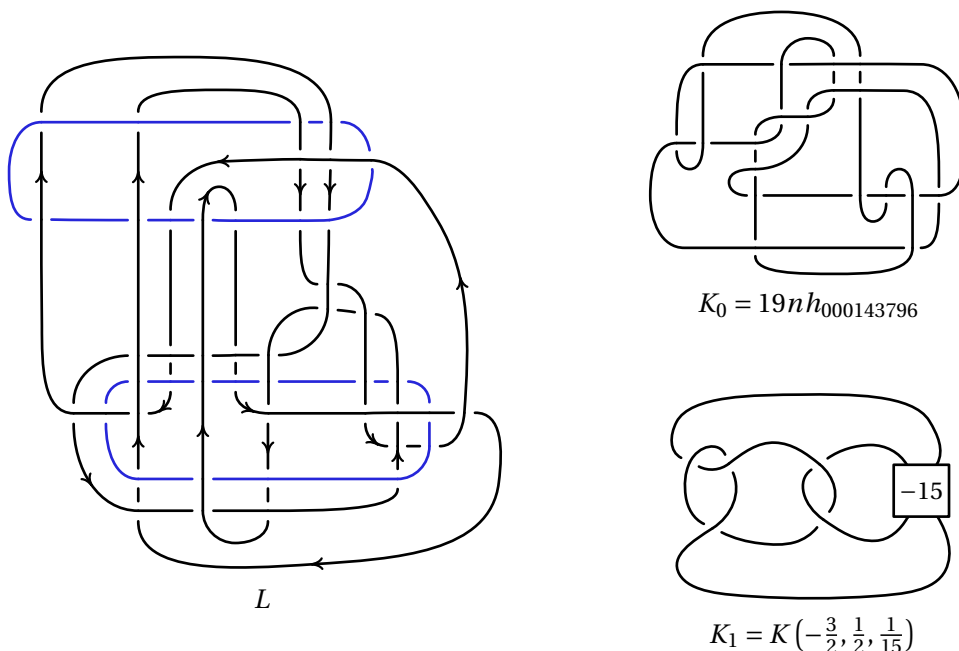


Figure 20. The links from Theorem 6.3. The standard projection of K_0 is shown top right. The knot K_1 is the Montesinos $K(-3/2, 1/2, 1/15)$, so the labeled box is filled in with 15 negative half-twists.

6 A few more non-slice knots

In this section, we deal with a few knots where the techniques in previous sections seem not to apply.

6.1 Smooth category. Recently, the 40-year-old question of whether the knot $17ns_{29}$ is smoothly slice was resolved:

6.2 Theorem [DKMPS]. *The $(2,1)$ -cable on the figure-8 knot, also known as $17ns_{29}$, is not smoothly slice.*

An alternate proof of this was given in [ACMPS] using genus bounds in definite 4-manifolds. In this section, we use the latter approach to show:

6.3 Theorem. *The knot $K_0 = 19nh_{000143796}$ is topologically slice but not smoothly slice.*

Proof of Theorem 6.3. First, the knot K_0 is topologically slice because $\Delta_{K_0} = 1$ [FQ, §11.7]. Let L and K_1 be the links shown in Figure 20. We denote the components of L as $L_0 \cup L_b \cup L_t$ where L_b and L_t are the bottom and top unknotted components, respectively. SnapPy [CDGW] easily checks the following or you can draw pictures: L_0 is isotopic to K_0 , and doing $+1$ Dehn surgery on both L_b and L_t turns L_0 into K_1 . Further, from Figure 20, we see that $lk(L_0, L_b) = 2$ and $lk(L_0, L_t) = 6$. Finally, the Seifert genus of K_1 is $g_3(K_1) = 9$ by computing knot Floer homology with [Sza]. The proof is now identical to that of Theorem 2 of [ACMPS], with K_0 playing the role of the $(2,1)$ -cable on the figure-8 knot and K_1 playing the role of the torus knot $T_{2,-19}$, but we give a sketch regardless.

Suppose $L_0 = K_0$ bounds a smooth disk D_0 in a 4-ball B_0 . Create a closed 4-manifold W by attaching $+1$ -framed 2-handles to B_0 along $L_b \cup L_t$ and capping the result off with another 4-ball B_1 . Since $L_b \cup L_t$ is the unlink, W is just $\mathbb{C}P^2 \# \mathbb{C}P^2$. Moreover, L_0 viewed from the other ball B_1 can be identified with K_1 after an isotopy. Let F_1 be a genus 9 Seifert surface for K_1 whose interior has been pushed into B_1 . Gluing F_0 and F_1 together along their common boundary gives a smooth closed surface $F \subset W$. Now $H_2(W) = \mathbb{Z}^2$, where the first generator is the core of the 2-handle attached to L_b together with a disk bounded by L_b in S^3 , and the second generator is the analogue with L_t . From the linking information about L , we see $[F] = (2, 6)$. However, by Corollary 1.7 of [Bry], any smooth connected surface representing the class $(2, 6)$ has genus at least 10, a contradiction. So K_0 is not smoothly slice. $\square$

6.4 Remark. We found the knot in Theorem 6.3 by a large-scale search for proofs similar to [ACMPS] in the spirit of our Section 5.11. That is, starting with a knot

K , we looked at closed geodesics in the hyperbolic structure of length at most 4, and recorded those giving unknots whose linking number with K was either 2 or 6. This gives candidates for L_b and L_t , and one can recover the diagram for the knot K' resulting from +1 surgery on L_b and L_t using [DOR]. The τ -invariant in knot Floer homology, computed via [Sza], was used to give a lower bound on the 4-ball genus of K' . Of the roughly 12,000 mystery knots where we tried this technique, $K_0 = 19nh_{000143796}$ was the only one where we succeeded.

6.5 Topological category. We will also need:

6.6 Theorem. *The nine alternating knots $K14a12741$, $16a350194$, $17ah_{0000055}$, $18ah_{0000122}$, $18ah_{3327857}$, $18ah_{4025786}$, $18ah_{4099296}$, $18ah_{4099297}$, and $19ah_{00000457}$ are not topologically slice.*

Proof. The five knots $K14a12741$, $16a350194$, $16a350194$, $17ah_{0000055}$, $18ah_{0000122}$, and $19ah_{00000457}$ are in fact the 2-bridge knots $K_{m^2, q}$ for (m^2, q) being $(25, 2)$, $(49, 4)$, $(121, 48)$, $(169, 48)$, and $(289, 190)$ respectively. We checked that the Casson-Gordon signatures obstruct each of these from being topologically slice, by directly calculating using the formulas from the original paper [CG2] and Theorem 3 therein.

For each of the four knots $18ah_{3327857}$, $18ah_{4025786}$, $18ah_{4099296}$, and $18ah_{4099297}$, we used the method of Section 2 to find a 1-band ribbon concordance from the initial knot K to $K_0 \# K_0$ for $K_0 = K7a1 = 7_7$. By Corollary 0.2 of [LN], the knot K_0 has infinite order in the topological concordance group $\mathcal{C}_1^{top}$. Hence, in $\mathcal{C}_1^{top}$ we have $[K] = 2[K_0] \neq 0$, and so K is not topologically slice. $\square$

7 Ribbon concordance graph

Our search for ribbon disks in Section 2 also produced many ribbon concordances between knots in $\mathcal{PS}_{19}$. In this section, we outline how these are used in the proof of Theorem 1.1, and how they relate to various open questions. Suppose K_0 and K_1 are knots in S^3 . Recall from [Gor] that a *ribbon concordance from K_1 to K_0* is a smooth annulus A embedded in $S^3 \times I$ with boundary components $K_0 \times \{0\}$ and $K_1 \times \{1\}$ so that the I -coordinate is a Morse function on A with no local maxima. In this case, we write $K_1 \geq K_0$. In particular, K is ribbon if and only if $K \geq \text{unknot}$. Agol recently proved that $\geq$ gives a partial order on isotopy classes of knots [Agol].

We will think of this partial order in terms of the *ribbon concordance graph* Γ_{RC} : the graph whose vertices are isotopy classes of knots with a *directed* edge from K_1 to K_0 if and only if $K_1 \geq K_0$. Note that K is ribbon if and only if there is a directed path starting at K and ending at the unknot. The connected components of Γ_{RC} as a topological space are exactly the concordance classes of (unoriented) knots [Gor,

§6]. In particular, the connected component of Γ_{RC} containing the unknot is the set of slice knots.

Considering the directed structure, define a *sink* for a component C of Γ_{RC} to be a vertex K so that every vertex $J \in C$ has a directed path from J to K . Because ribbon concordance is a partial order, a component can have at most one sink. The ribbon-slice conjecture is that the unknot is the unique sink of its component, and Gordon asked in [Gor] whether every component of Γ_{RC} has a sink.

Our data allows us to partially reconstruct the subgraph of Γ_{RC} whose vertices are $\mathcal{PS}_{19} \cup \{\text{unknot}\}$. Let Γ_0 be the directed graph whose vertices are $\mathcal{PS}_{19} \cup \{\text{unknot}\}$ with an edge from K_1 to K_0 if we found a ribbon concordance $K_1 \geq K_0$. We define $\Gamma_{\mathcal{PS}_{19}}$ to be the subgraph of Γ_0 where we remove all the isolated vertices, that is, all the knots that are not involved in any ribbon concordance we found. We call $\Gamma_{\mathcal{PS}_{19}}$ the *partial ribbon concordance graph* of $\mathcal{PS}_{19}$. (Technical aside: elements of $\mathcal{PS}_{19}$ are not isotopy classes of knots but rather defined up to a possibly orientation-reversing homeomorphism of S^3 . So $\Gamma_{\mathcal{PS}_{19}}$ is a subgraph not of Γ_{RC} but of a $\mathbb{Z}/2\mathbb{Z}$ -quotient of it.)

7.1 Properties of the partial concordance graph. The graph $\Gamma_{\mathcal{PS}_{19}}$ has 1,677,696 vertices and 1,835,130 edges. Of the edges, some 1,420,528 end at the unknot. As a topological space, it has 524 connected components. The largest component U is the one containing the unknot. The other knots in U are exactly the 1,633,786 ribbon knots we identified in Theorem 2.1; the unknot is the sink of U as predicted by the slice-ribbon conjecture. A very small piece of U is shown in Figures 21, 22, and 23.

The other components of $\Gamma_{\mathcal{PS}_{19}}$ have mean size 84.0 and median size 47.0; the largest such component has size 1673, with sink the Conway knot $K11n34$. Figure 24 shows the sizes roughly follow a power-law distribution. In contrast with Figure 21, every one of these other components has the following simple structure as a directed graph: there is a sink and every edge ends there. Note this is consistent with the generalized slice-ribbon conjecture that every component of Γ_{RC} has a sink.

It is natural to ask how various invariants behave with respect to ribbon concordance.

- (1) When the exterior $S^3 \setminus K$ is hyperbolic, i.e. has a complete finite-volume Riemannian metric of constant curvature -1 , we define $\text{vol}(K)$ to be that volume. This notion naturally extends to all knots by considering the Gromov norm; here, unknots and torus knots have volume 0. Gordon asked in [Gor] whether $K_1 \geq K_0$ implies $\text{vol}(K_1) \geq \text{vol}(K_0)$. This is true for $\Gamma_{\mathcal{PS}_{19}}$ in a very strong form. For $K_1 > K_0$, the smallest value of $\text{vol}(K_1) - \text{vol}(K_0)$ was about 3.16, occurring for $K_1 = K6a3$ and $K_0 = \text{unknot}$. Moreover, the largest ratio $(\text{vol}(K_1) - \text{vol}(K_0)) / \text{vol}(K_1)$ was ≈ 0.74 for $K_1 = 19ah_{35125143}$ and $K_0 = K14a14913$; that is, the volume always declined by more than 25% along each edge of $\Gamma_{\mathcal{PS}_{19}}$.

- (2) Conjecture 9 of [GO] posits that if $K_1 \geq K_0$ and K_1 is alternating then K_0 must be as well. Equivalently, there are no edges in Γ_{RC} from an alternating knot to a nonalternating knot. The graph $\Gamma_{\mathcal{PS}_{19}}$ has 1,484,676 edges where one end is an alternating knot and the other is not. In every case, these go in the expected direction.
- (3) Zemke showed that if $K_1 \geq K_0$ then the induced map on knot Floer homology $\widehat{HFK}(K_0) \rightarrow \widehat{HFK}(K_1)$ is injective. In particular, for each (i, j) -bigrading one has $\dim \widehat{HFK}_i(K_1, j) \geq \dim \widehat{HFK}_i(K_0, j)$ as $\mathbb{F}_2$ -vector spaces [Zem]. (The analogous result holds for Khovanov homology [LZ].) Given this, a natural question is whether one could have $\widehat{HFK}(K_0) \cong \widehat{HFK}(K_1)$ when $K_1 > K_0$. For the total dimension of $\widehat{HFK}$, the smallest absolute drop was $K_1 = K6a3$ and $K_0 = \text{unknot}$ where it declined from 9 to 1. The smallest decline as a proportion was the edge from $16n30703$ to $K6a3$ where it decreased by some 78.0%. (In comparing this decline with that of hyperbolic volume, recall the connection shown in Figure 3).
- (4) A consequence of [Zem] is that the Seifert genus respects ribbon concordance: if $K_1 \geq K_0$ then $g_3(K_1) \geq g_3(K_0)$. For $\Gamma_{\mathcal{PS}_{19}}$, moreover, every edge strictly reduced g_3 . This prompts us to ask: are there prime knots $K_1 > K_0$ with the same g_3 ?
- (5) By [Miy, Proposition 5], if $K_1 \geq K_0$ and K_1 is fibered then so is K_0 . Given this, it is natural to ask whether the dimension of $\widehat{HFK}$ in the top Alexander grading is non-increasing. This is true for 99.25% of the edges in $\Gamma_{\mathcal{PS}_{19}}$, but there are 3,094 cases where it increases.

7.2 Propagating information. Finally, we use $\Gamma_{\mathcal{PS}_{19}}$ to complete the proof of the main theorem. The idea is that, since each component C of $\Gamma_{\mathcal{PS}_{19}}$ consists of smoothly concordant knots, if we know something about one K in C , we can perhaps deduce the same about every knot in C . While the smooth obstructions of Section 4 are all concordance invariants, and so apply or not to every element in C uniformly, this is not the case for the techniques in Sections 3, 5, and 6.

Proof of Theorem 1.9. Of necessity, we only give an outline and defer to [DG] for the detailed data.

We first do the topological category. Recall from Theorem 2.1 that we identified 1,633,786 ribbon knots in $\mathcal{PS}_{19}$. Moreover, $\mathcal{PS}_{19}$ contains 34,930 knots where $\Delta_K = 1$, which are all topologically slice by [FQ, §11.7]; this contributes 18,954 topologically slice knots beyond the known ribbon knots. Any component C of $\Gamma_{\mathcal{PS}_{19}}$ containing a topologically slice knot consists entirely of such knots. For example, the Conway knot $K11n34$ has $\Delta_K = 1$ and it is the sink of a component C of size 1673; this tells

sink	#C	sink	#C	sink	#C
$K13n65$	199	$K13n3871$	88	$K14n4621$	56
$K13n4582$	138	$K13n3897$	86	$K14n9023$	43
$K13n3872$	127	$K14n18911$	59	$K14n11063$	43
$K13n3936$	127	$K14n18909$	59	$K14n4425$	37
$K14n21673$	99	$K14n3713$	58	$K14n5486$	33

Table 12. The 15 largest components C of $\Gamma_{\mathcal{PS}_{19}}$ whose smooth slice status is unknown; together they contain 1251 knots, some 11.0% of the those in $\mathcal{PS}_{19}$ whose smooth slice status is unknown. With the exceptions of $K14n3713$ and $K14n4621$, all of the sink knots listed above have $\Delta_K = 1$ and hence those components consist entirely of topologically slice knots. The components with sinks $K14n3713$ and $K14n4621$ are the only two in $\mathcal{PS}_{19}$ where we were unable to determine whether the knots are topologically slice.

us 1672 knots with $\Delta_K \neq 1$ are topologically slice. There are 17 other components whose sink is a knot with $\Delta_K = 1$. Collectively, these 18 components contribute 3457 more topologically slice knots. This gives the claimed total of at least 1,656,198 topologically slice knots in $\mathcal{PS}_{19}$.

Theorem 3.1 gives us 2,211,761 knots in $\mathcal{PS}_{19}$ that are not topologically slice, and Theorem 6.6 gives 9 more. We can also propagate this information via $\Gamma_{\mathcal{PS}_{19}}$. For example, the component C of $\Gamma_{\mathcal{PS}_{19}}$ with sink $K14n5389$ has $\#C = 25$. The method of Section 3.20 with $(m, q) = (2, 3)$ shows 13 of the knots K cannot be topologically slice and hence the other knots in C are not as well. (The 13 knots where the method works have $H_1(B_2; \mathbb{Z}) = \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/27\mathbb{Z}$ whereas the other 12 have $H_1(B_2; \mathbb{Z})$ one of $(\mathbb{Z}/3\mathbb{Z})^2$, $(\mathbb{Z}/9\mathbb{Z})^2$, or $(\mathbb{Z}/3\mathbb{Z})^2 \oplus \mathbb{Z}/9\mathbb{Z}$. Note that Lemma 3.7 gives only two possible metabolizers for $\mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/27\mathbb{Z}$ whereas there are 13 possibilities for $\mathbb{Z}/9\mathbb{Z} \oplus \mathbb{Z}/9\mathbb{Z}$, which makes it harder to rule them all out.) This kind of deduction applies to some 144 knots, for a total of at least 2,211,914 knots in $\mathcal{PS}_{19}$ that are not topologically slice. This completes our discussion in the topological category.

In the smooth setting, Theorem 4.1 obstructs smooth slicing of 10,618 knots beyond those known not to be topologically slice. As the obstructions in Theorem 4.1 are concordance invariants, we cannot propagate them via $\Gamma_{\mathcal{PS}_{19}}$. Theorem 5.10 gives 25 additional knots that are not smoothly slice. Two of these, $K11n34$ and $K13n866$, are sinks of components of $\Gamma_{\mathcal{PS}_{19}}$ of sizes 1673 and 142 respectively, which adds 1813 nonslice knots. Theorem 6.2 and Theorem 6.3 contribute one nonslice knot apiece, for a total of at least 2,224,372 knots in $\mathcal{PS}_{19}$ that are not smoothly slice. $\square$

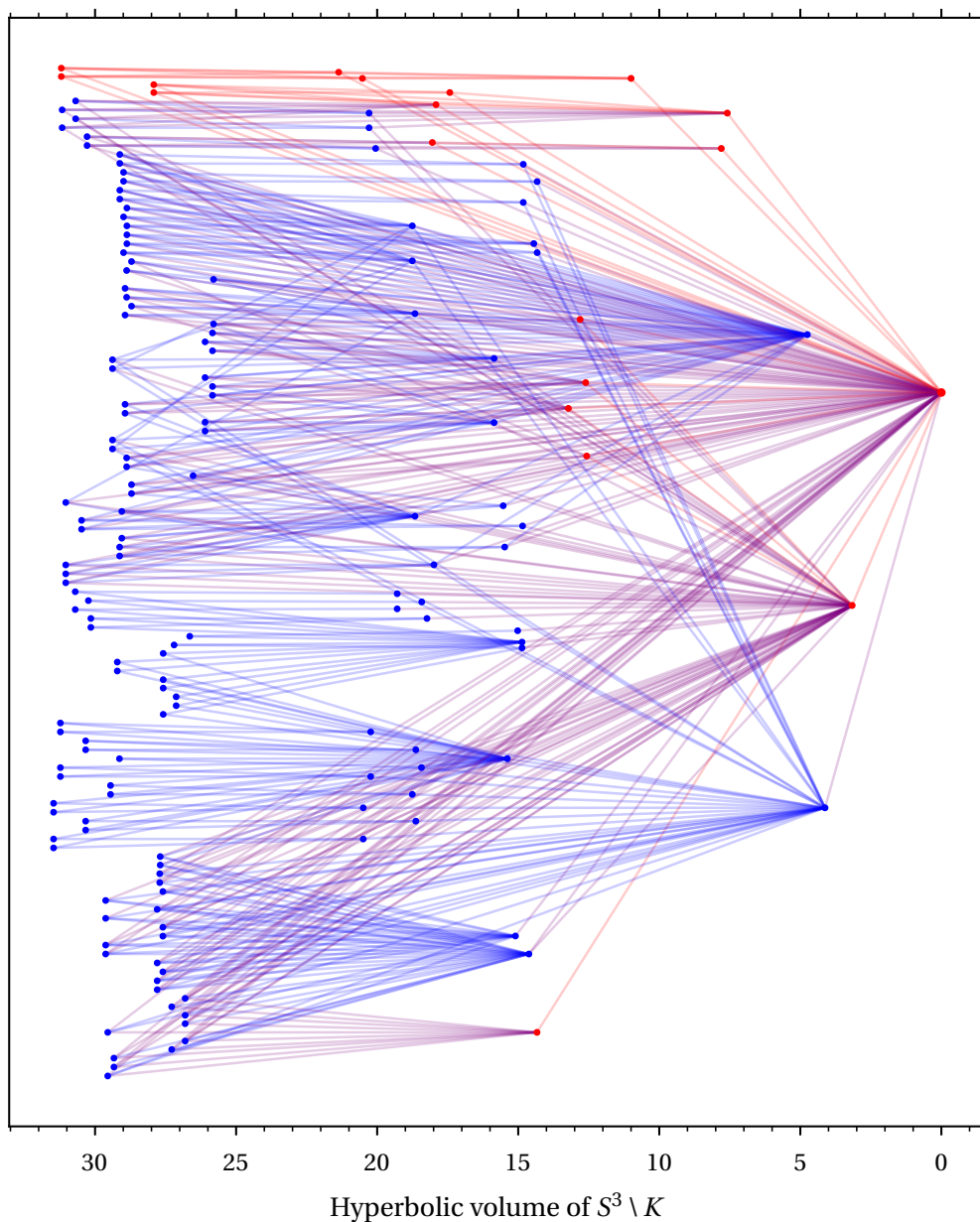


Figure 21. A small part of the component of $\Gamma_{\mathcal{P}_{S_{19}}}$ containing the unknot. Each knot is indicated by a dot, where alternating knots are red and other knots are blue. Only edges corresponding to known 1-band ribbon cobordisms are drawn; in each case, they are directed so that the volume decreases, and their color is the average of that of their endpoints. The 168 knots used here were carefully chosen to make the picture complicated and interesting. The vertical coordinates of the dots were algorithmically selected to make the picture more comprehensible; they do not have an intrinsic meaning. See also Figure 22 and Figure 23.

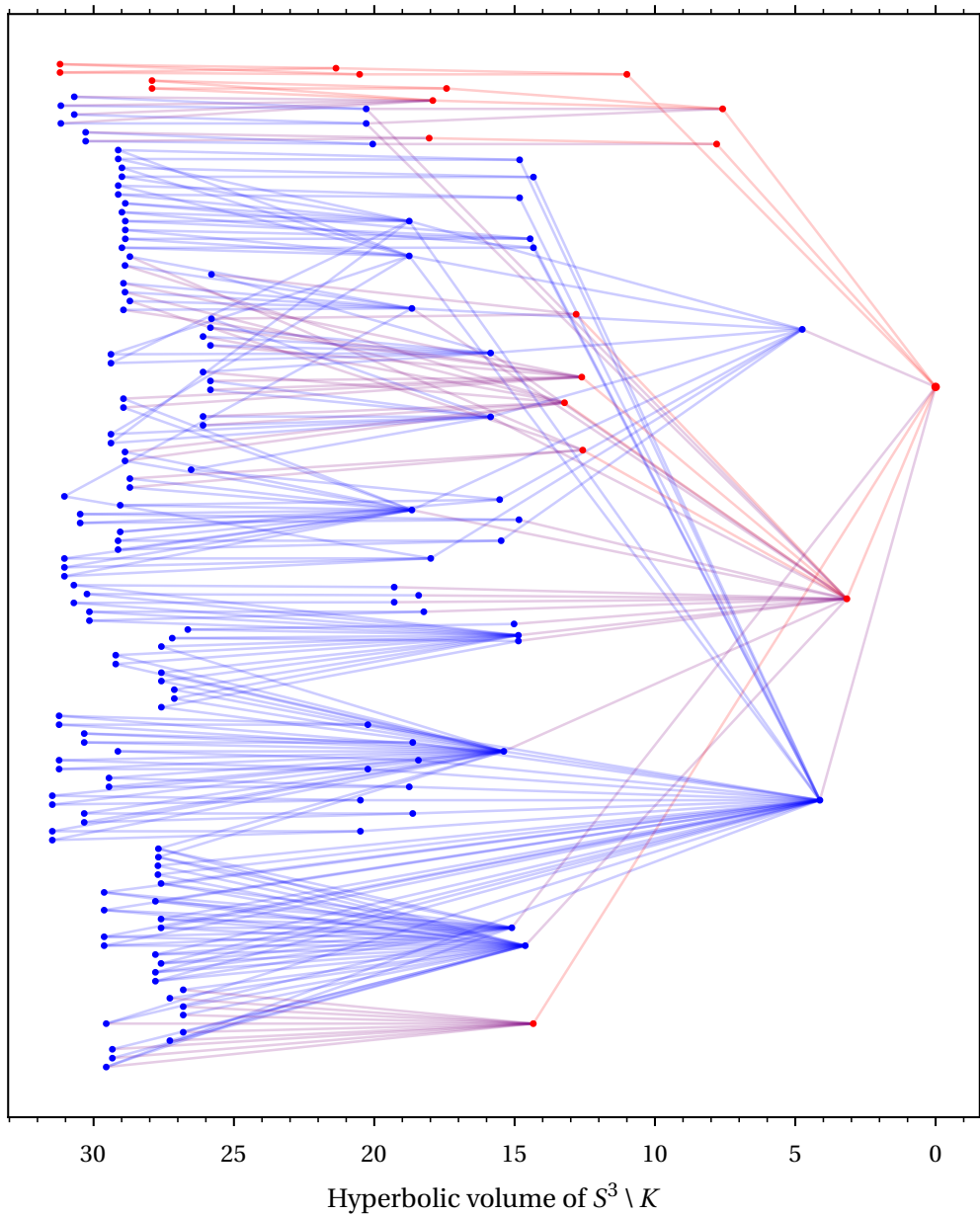


Figure 22. A subgraph of $\Gamma_{\mathcal{PS}_{19}}$ with the same vertices as that in Figure 21, but with a minimal subset of edges generating the underlying partial order.

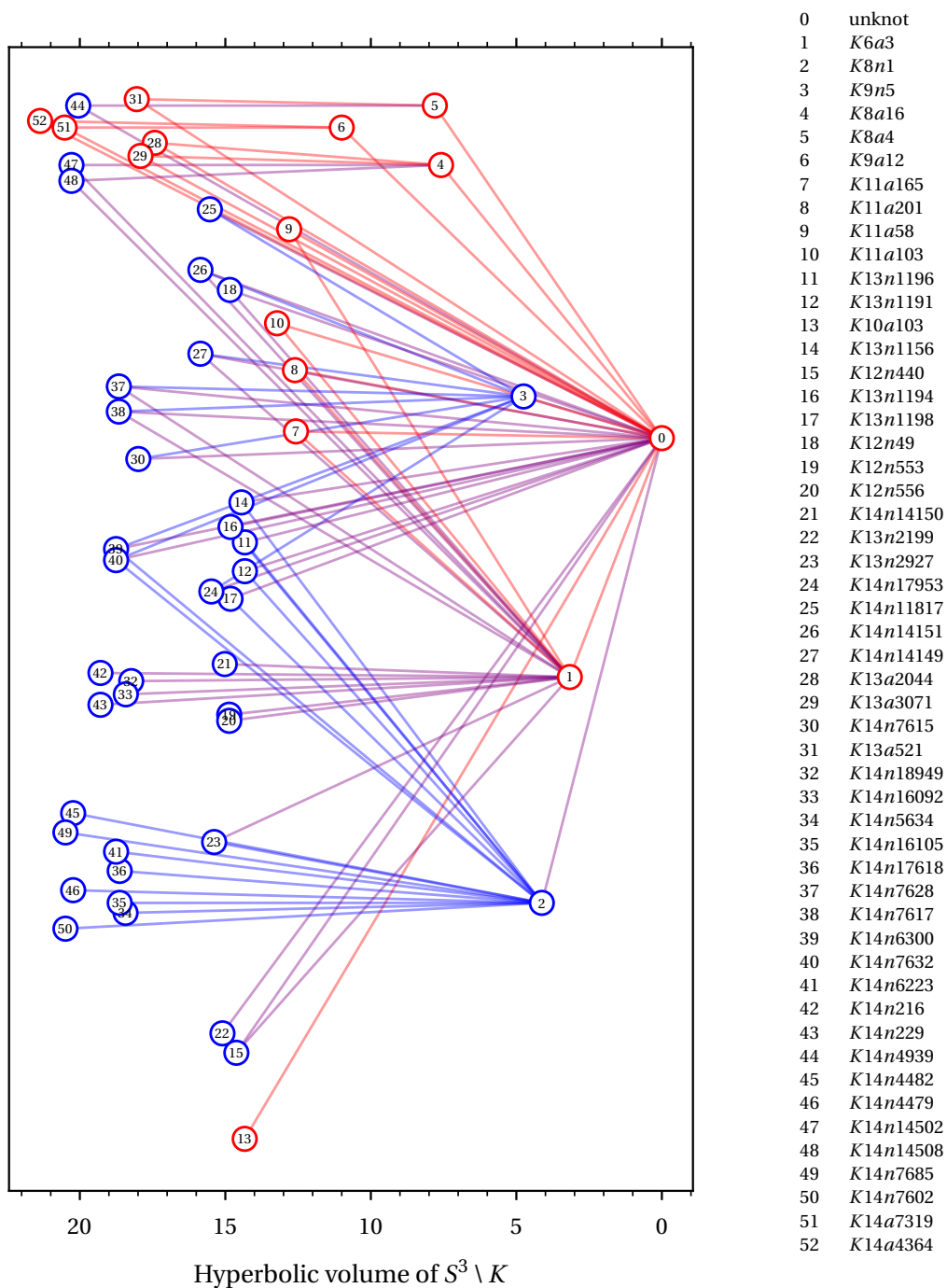


Figure 23. The subgraph of the one shown in Figure 21 corresponding to knots with at most 17 crossings, which in this case is equivalent to $\text{vol}(S^3 \setminus K) < 22$.

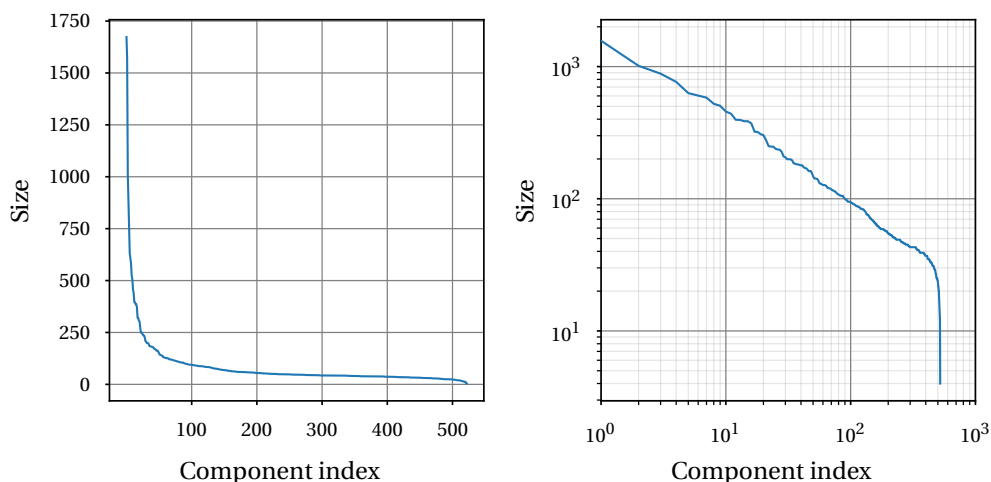


Figure 24. This pair of plots shows the sizes of the components of $\Gamma_{\mathcal{P}_{S_{19}}}$ that do not contain the unknot. On the horizontal axis, the 523 components are ordered from largest to smallest. The two plots differ only in the use of log-scales on the right so that the power-law behavior is evident.

References

- [AKW] T. Abe, M. Kegel, and N. Weiss. Complexity of equal 0-surgeries. Preprint 2025, 23 pages. [arXiv:2401.06015](#).
- [ACMPS] P. Aceto, N. A. Castro, M. Miller, J. Park, and A. Stipsicz. Slice obstructions from genus bounds in definite 4-manifolds, 2023. 6 pages. [arXiv:2303.10587](#).
- [Agol] I. Agol. [Ribbon concordance of knots is a partial ordering](#). *Commun. Am. Math. Soc.* **2** (2022), 374–379. [arXiv:2201.03626](#), [MR4520779](#).
- [AST] I. Agol, P. A. Storm, and W. P. Thurston. [Lower bounds on volumes of hyperbolic Haken 3-manifolds](#). *J. Amer. Math. Soc.* **20** (2007), 1053–1077. With an appendix by Nathan Dunfield. [arXiv:0506338](#), [MR2328715](#).
- [ABDE⁺] T. Applebaum, S. Blackwell, A. Davies, T. Edlich, A. Juhász, M. Lackenby, N. Tomašev, and D. Zheng. The unknotting number, hard unknot diagrams, and reinforcement learning. *Exp. Math.* (to appear). [arXiv:2409.09032](#).
- [Asc] M. Aschbacher. [Finite group theory](#), volume 10 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, second edition, 2000. [MR1777008](#).
- [BCP] W. Bosma, J. Cannon, and C. Playoust. [The Magma algebra system. I. The user language](#). *J. Symbolic Comput.* **24** (1997), 235–265. Computational algebra and

- number theory (London, 1993). <http://doi.org/10.1006/jSCO.1996.0125> MR1484478.
- [Bry] J. Bryan. *Seiberg-Witten theory and $\mathbb{Z}/2^p$ actions on spin 4-manifolds*. *Math. Res. Lett.* **5** (1998), 165–183. MR1617929.
- [BB] R. Budney and B. A. Burton. *Embeddings of 3-manifolds in S^4 from the point of view of the 11-tetrahedron census*. *Exp. Math.* **31** (2022), 988–1013. [arXiv:0810.2346](https://arxiv.org/abs/0810.2346), MR4477418.
- [Bur] B. A. Burton. *The next 350 million knots*. In *36th International Symposium on Computational Geometry*, volume 164 of *LIPIcs. Leibniz Int. Proc. Inform.*, pages Art. No. 25, 17. Schloss Dagstuhl. Leibniz-Zent. Inform., Wadern, 2020. MR4117738.
- [BBP⁺] B. A. Burton, R. Budney, W. Pettersson, et al. *Regina: Software for low-dimensional topology, 1999–2023*. <http://regina-normal.github.io>
- [CG1] A. J. Casson and C. M. Gordon. *A loop theorem for duality spaces and fibred ribbon knots*. *Invent. Math.* **74** (1983), 119–137. MR722728.
- [CG2] A. J. Casson and C. M. Gordon. *Cobordism of classical knots*. In *À la recherche de la topologie perdue*, volume 62 of *Progr. Math.*, pages 181–199. Birkhäuser Boston, Boston, MA, 1986. With an appendix by P. M. Gilmer. MR900252.
- [CL] A. J. Casson and D. D. Long. *Algorithmic compression of surface automorphisms*. *Invent. Math.* **81** (1985), 295–303. MR799268.
- [Coh] H. Cohen. *A course in computational algebraic number theory*, volume 138 of *Graduate Texts in Mathematics*. Springer-Verlag, Berlin, 1993. MR1228206.
- [CDGW] M. Culler, N. M. Dunfield, M. Goerner, and J. R. Weeks. *SnapPy, a computer program for studying the geometry and topology of 3-manifolds, Version 3.2, 2025*. <https://snappy.computop.org>
- [DKMPS] I. Dai, S. Kang, A. Mallick, J. Park, and M. Stoffregen. *The $(2, 1)$ -cable of the figure-eight knot is not smoothly slice*. *Invent. Math.* (to appear). [arXiv:2207.14187](https://arxiv.org/abs/2207.14187).
- [DJLTs] A. Davies, A. Juhász, M. Lackenby, and N. Tomašev. *The signature and cusp geometry of hyperbolic knots*. *Geom. Topol.* **28** (2024), 2313–2343. [arXiv:2111.15323](https://arxiv.org/abs/2111.15323), MR4793642.
- [DF] D. S. Dummit and R. M. Foote. *Abstract algebra*. John Wiley & Sons, Inc., Hoboken, NJ, third edition, 2004. MR2286236.
- [Dun] N. M. Dunfield. *A census of exceptional Dehn fillings*. In *Characters in low-dimensional topology*, volume 760 of *Contemp. Math.*, pages 143–155. Amer. Math. Soc., 2020. 1812.11940, MR4193924.

- [DG] N. M. Dunfield and S. Gong. [Ribbon concordances and slice obstructions: code and data](#). Harvard Dataverse, 2025.
- [DLS] N. M. Dunfield, R. Lipshitz, and D. Schütz. Local equivalence and refinements of Rasmussen’s s-invariant. *J. Topol.* (to appear). 47 pages. [arXiv:2312.09114](#).
- [DOR] N. M. Dunfield, M. Obeidin, and C. G. Rudd. [Computing a link diagram from its exterior](#). *Discrete Comput. Geom.* **71** (2024), 121–159. [2112.03251](#), [MR4685711](#).
- [FM] P. Feller and D. McCoy. [On 2-bridge knots with differing smooth and topological slice genera](#). *Proc. Amer. Math. Soc.* **144** (2016), 5435–5442. [arXiv:1508.01431](#), [MR3556284](#).
- [Fox] R. H. Fox. Some problems in knot theory. In *Topology of 3-manifolds and related topics (Proc. The Univ. of Georgia Institute, 1961)*, pages 168–176. Prentice-Hall, Inc., Englewood Cliffs, NJ, 1961. [MR140100](#).
- [FQ] M. H. Freedman and F. Quinn. *Topology of 4-manifolds*, volume 39 of *Princeton Mathematical Series*. Princeton University Press, Princeton, NJ, 1990. [MR1201584](#).
- [FV] S. Friedl and S. Vidussi. [A survey of twisted Alexander polynomials](#). In *The mathematics of knots*, volume 1 of *Contrib. Math. Comput. Sci.*, pages 45–94. Springer, Heidelberg, 2011. https://doi.org/10.1007/978-3-642-15637-3_3 [arXiv:0905.0591](#), [MR2777847](#).
- [GAP] The GAP Group. *GAP – Groups, Algorithms, and Programming, Version 4.13.1*, 2024. <https://www.gap-system.org>
- [Gor] C. M. Gordon. [Ribbon concordance of knots in the 3-sphere](#). *Math. Ann.* **257** (1981), 157–170. [MR634459](#).
- [GO] J. E. Greene and B. Owens. Alternating links, rational balls, and cube tilings. *J. Eur. Math. Soc. (JEMS)*. [arXiv:2212.06248](#).
- [GHMR] S. Gukov, J. Halverson, C. Manolescu, and F. Ruehle. Searching for ribbons with machine learning, 2023. preprint, 23 pages. [arXiv:2304.09304](#).
- [HKL] C. Herald, P. Kirk, and C. Livingston. [Metabelian representations, twisted Alexander polynomials, knot slicing, and mutation](#). *Math. Z.* **265** (2010), 925–949. [MR2652542](#).
- [HW] C. D. Hodgson and J. R. Weeks. Symmetries, isometries and length spectra of closed hyperbolic three-manifolds. *Experiment. Math.* **3** (1994), 261–274. <http://projecteuclid.org/euclid.em/1048515809> [MR1341719](#).
- [HR] D. Holt and G. Royle. [A census of small transitive groups and vertex-transitive graphs](#). *J. Symbolic Comput.* **101** (2020), 51–60. [arXiv:1811.09015](#), [MR4109707](#).

- [JMZ] A. Juhász, M. Miller, and I. Zemke. [Knot cobordisms, bridge index, and torsion in Floer homology](#). *J. Topol.* **13** (2020), 1701–1724. [arXiv:1904.02735](#), [MR4186142](#).
- [Kaw] A. Kawauchi. *A survey of knot theory*. Birkhäuser Verlag, Basel, 1996. Translated and revised from the 1990 Japanese original by the author. [MR1417494](#).
- [Kho] M. Khovanov. [Patterns in knot cohomology. I. Experiment. Math.](#) **12** (2003), 365–374. [arXiv:0201306](#), [MR2034399](#).
- [KL] P. Kirk and C. Livingston. [Twisted Alexander invariants, Reidemeister torsion, and Casson-Gordon invariants](#). *Topology* **38** (1999), 635–661. [MR1670420](#).
- [Kup] G. Kuperberg. [Algorithmic homeomorphism of 3-manifolds as a corollary of geometrization](#). *Pacific J. Math.* **301** (2019), 189–241. [MR4007377](#).
- [LZ] A. S. Levine and I. Zemke. [Khovanov homology and ribbon concordances](#). *Bull. Lond. Math. Soc.* **51** (2019), 1099–1103. [arXiv:1903.01546](#), [MR4041014](#).
- [LM] L. Lewark and D. McCoy. [On calculating the slice genera of 11- and 12-crossing knots](#). *Exp. Math.* **28** (2019), 81–94. [arXiv:1508.01098](#), [MR3938580](#).
- [Lic] W. B. R. Lickorish. *An introduction to knot theory*, volume 175 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1997. [MR1472978](#).
- [LS] R. Lipshitz and S. Sarkar. [A refinement of Rasmussen’s S-invariant](#). *Duke Math. J.* **163** (2014), 923–952. [arXiv:1206.3532](#), [MR3189434](#).
- [Lis] P. Lisca. [Lens spaces, rational balls and the ribbon conjecture](#). *Geom. Topol.* **11** (2007), 429–472. [arXiv:0701610](#), [MR2302495](#).
- [Liv] C. Livingston. [The algebraic concordance order of a knot](#). *J. Knot Theory Ramifications* **19** (2010), 1693–1711. [MR2755496](#).
- [LN] C. Livingston and S. Naik. [Obstructing four-torsion in the classical knot concordance group](#). *J. Differential Geom.* **51** (1999), 1–12. <http://projecteuclid.org/euclid.jdg/12144250231> [arXiv:9802071](#), [MR1703602](#).
- [MP] C. Manolescu and L. Piccirillo. [From zero surgeries to candidates for exotic definite 4-manifolds](#). *J. Lond. Math. Soc. (2)* **108** (2023), 2001–2036. [arXiv:2102.04391](#), [MR4668522](#).
- [Miy] K. Miyazaki. [A note on genera of band sums that are fibered](#). *J. Knot Theory Ramifications* **27** (2018), 1871002, 3. [MR3876350](#).
- [Nak] K. Nakamura. [Trace embeddings from zero surgery homeomorphisms](#). *J. Topol.* **16** (2023), 1641–1664. [arXiv:2203.14270](#), [MR4828048](#).

-
- [NZ] W. D. Neumann and D. Zagier. [Volumes of hyperbolic three-manifolds](#). *Topology* **24** (1985), 307–332. [MR815482](#).
 - [Nos] T. Nosaka. Cellular chain complexes of universal covers of some 3-manifolds. *J. Math. Sci. Univ. Tokyo* **29** (2022), 89–113. [arXiv:2107.08851](#), [MR4414248](#).
 - [OS] B. Owens and F. Swenton. [An Algorithm to Find Ribbon Disks for Alternating Knots](#). *Experimental Mathematics* **0** (2023), 1–19. [arXiv:2102.11778](#).
 - [Pic] L. Piccirillo. [The Conway knot is not slice](#). *Ann. of Math. (2)* **191** (2020), 581–591. [arXiv:1808.02923](#), [MR4076631](#).
 - [Ras] J. Rasmussen. [Khovanov homology and the slice genus](#). *Invent. Math.* **182** (2010), 419–447. [arXiv:0402131](#), [MR2729272](#).
 - [Ren] Q. Ren. [Lee filtration structure of torus links](#). *Geom. Topol.* **28** (2024), 3935–3960. [arXiv:2305.16089](#), [MR4843752](#).
 - [Rub] D. Ruberman. Fibered example of topologically slice knots. MathOverflow Answer. <https://mathoverflow.net/q/215495>
 - [Sar] S. Sarkar. [Ribbon distance and Khovanov homology](#). *Algebr. Geom. Topol.* **20** (2020), 1041–1058. [MR4092319](#).
 - [SSS] S. Sarkar, C. Scaduto, and M. Stoffregen. [An odd Khovanov homotopy type](#). *Adv. Math.* **367** (2020), 107112, 51. [MR4078823](#).
 - [Saw] W. Sawin. Bilinear forms on abelian p -groups: Images of weak metabolizers in the Frattini quotient. MathOverflow Answer. <https://mathoverflow.net/q/473984>
 - [Sch1] D. Schütz. [A scanning algorithm for odd Khovanov homology](#). *Algebr. Geom. Topol.* **22** (2022), 1287–1324. [arXiv:2008.07410](#), [MR4474786](#).
 - [Sch2] D. Schütz. KnotJob (computer software), gray version, 2023. <https://www.maths.dur.ac.uk/users/dirk.schuetz/knotjob.html>
 - [Sch3] D. Schütz. A note on the X-torsion order of a knot, 2024. Preprint 2024, 6 pages. [arXiv:2412.05156](#).
 - [Sch4] D. Schütz. [On an integral version of the Rasmussen invariant](#). *Michigan Math. J.* **75** (2025), 65–88. [arXiv:2202.00445](#), [MR4873797](#).
 - [Sch5] D. Schütz. On Variations of s-invariants from $\mathrm{sl}(3)$ -link homology. Preprint 2025, 29 pages. [arXiv:2504.07806](#).
 - [Sch6] D. Schütz. Two second Steenrod squares for odd Khovanov homology. Preprint 2022, 38 pages. [arXiv:2209.00389](#).

- [Sto] A. Stoimenow. [Some inequalities between knot invariants](#). *Internat. J. Math.* **13** (2002), 373–393. [MR1911864](#).
- [Sza] Z. Szabó. Knot Floer homology calculator, 2017. <https://web.math.princeton.edu/~szabo/HFKcalc.html>
- [Tei] P. Teichner. Slice knots: Knot theory in the 4th dimension, 2011. Lecture notes by Justin Roberts, Julia Collins and Mark Powell. <https://people.mpim-bonn.mpg.de/teichner/Math/surveys.html>
- [Sage] The Sage Developers. *SageMath, the Sage Mathematics Software System (Version 10.0)*, 2023. <https://www.sagemath.org>
- [Thi] M. B. Thistlethwaite. [The enumeration and classification of prime 20-crossing knots](#). *Algebr. Geom. Topol.* **25** (2025), 329–344. [MR4877260](#).
- [Zem] I. Zemke. [Knot Floer homology obstructs ribbon concordance](#). *Ann. of Math. (2)* **190** (2019), 931–947. [arXiv:1902.04050](#), [MR4024565](#).